
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

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ABSTRACT

Algorithms now function as pervasive information mediators, spanning social-media feeds, targeted advertising, and the rapidly widening deployment of large language models (LLMs). Such mediation induces a coupled process in which each agent fuses a private, algorithmically shaped signal with observational learning from the actions of its predecessors before choosing how to act. We formalize this process as *controlled sequential social learning*, in which an information-mediating planner—an LLM, for instance—governs the information structure presented to a sequence of agents who simultaneously learn from one another’s decisions. We analyze the two canonical objectives that bracket the space of planner incentives: maximizing social welfare (an *altruistic* planner) and inducing a fixed, self-serving action (a *biased* planner). The framework defines a new class of social-learning optimization problems, coupling the planner’s dynamic program to the decentralized, utility-maximizing action choices and Bayesian belief updates of the agents. Within this setting we prove that the altruistic value function is convex in the public belief and, on that foundation, derive a complete characterization of the optimal policies of both planners, each of which resolves the trade-off between the cost of information provision and the payoff of the agent choices it induces. The optimal planner operates in sharply delineated regimes indexed by the public belief—investing the maximum admissible resource, withholding investment entirely, or raising or lowering investment monotonically in the belief—and, over an identifiable range of beliefs, the biased planner deliberately *obfuscates* agents’ signals to suppress unfavorable information. Even under stringent transparency constraints—information parity with individuals, no fabrication or cherry-picking, and full observability of the planner’s actions—information mediation alone can move social welfare substantially in either direction. Controlled simulations in which LLMs instantiate both planner and agents corroborate this analysis: the LLM planner adopts policies that closely track the non-trivial optima our theory predicts, and its interpretable deviations align with the non-Bayesian updating documented in both human subjects and LLMs trained on human-generated data—evidence that the characterization is robust to the departures from rationality we observe. Together, these results establish controlled social learning as a tractable analytical foundation, and a concrete testbed, for studying—and ultimately regulating—LLM information mediators.

1 INTRODUCTION

Large Language Models (LLMs) are increasingly entrusted with socially critical mediation functions, ranging from search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024) to personal assistants that dispense medical and political advice (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023). This capability is inseparable from a documented capacity for influence: LLMs have been shown to be persuasive (Potter et al., 2024; Carrasco-Farre, 2024; Rogiers et al., 2024), manipulative (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategic (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b) in their interactions with individual users. When exercised at scale, however, such influence never operates in isolation. It is superimposed on the organic diffusion of information through which people observe and learn from one another, producing coupled social dynamics that can amplify or attenuate a

model’s impact. It is this coupling, rather than any single interaction, that governs the aggregate societal effect—and yet it is precisely this coupling that the prevailing focus on one-on-one interaction leaves unexamined.

To make this concrete, consider an LLM-powered recommendation system that presents a new restaurant to a sequence of prospective customers. The quality of the restaurant—the payoff-relevant state of the world—is either good or bad and is unknown to users and system alike. Each user receives a personalized recommendation (a private signal) and then decides whether to patronize the restaurant. The recommendation system—the *planner*, in our terminology—can expend resources to render that signal more informative, for instance by having the LLM generate a review finely tailored to the user’s known preferences. The expenditure carries a concrete cost, reflecting the market research, user surveys, and data acquisition that tailoring information effectively demands.

A user’s decision, however, is never a function of the private signal alone: each user also observes the actions of its predecessors—should the previous ten users have chosen to patronize the restaurant, the next user rationally infers that they likely received favorable information. This mechanism, *social learning*, is a decisive force in the formation of human opinion across consequential domains, including the adoption of vaccines (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)), the diffusion of new technology (e.g. Gillingham & Bollinger (2021); Weber (2012)), and the evolution of political and moral attitudes (e.g., Brady et al. (2021); Guilbeault et al. (2018)). The central object of this work is the strategic interaction between a planner who controls the private information supplied to agents and a population that simultaneously engages in social learning. Under this coupling the planner must reason not merely about the current user but about the information *externality* its choices impose—an externality that reshapes the public belief confronting every future user, and thereby the entire downstream learning trajectory.

Although real-world interactions are immensely complex, we distill this dynamic into a tractable formal model of *controlled social learning* in which an algorithmic planner strategically selects, at a cost, the precision of the private signal delivered to each agent in a sequence. The planner’s leverage is deliberately circumscribed and subtle: it never falsifies information, choosing only how much to invest in rendering each signal clear and informative. The paradigm poses three questions that our analysis answers directly: how a planner can wield this constrained instrument to steer collective beliefs, how its optimal strategy adapts to the evolving state of public opinion, and what consequences follow for social welfare. We answer all three within the model, in closed form for the theory and through controlled LLM simulation for the empirical claims.

These answers hinge on the planner’s objective, and we study the two canonical cases that bracket the space of incentives. An *altruistic* planner, exemplified by an ideal recommendation system, seeks to maximize user utility by inducing the correct choice—patronizing the restaurant if and only if it is good. A *biased* planner—one receiving, say, a kickback from the restaurant—instead seeks to induce a specific action, patronage, irrespective of the true state.

This dichotomy is related to, but distinct from, the notion of alignment. The altruistic objective is by construction aligned with maximizing each agent’s *expected utility* under state uncertainty. The biased objective, being state-independent, is not so aligned; yet its influence is not invariably detrimental. When the true state happens to favor the planner’s preferred action—when the restaurant is in fact good—the biased planner’s influence coincides with the agent’s *realized utility* in that instance. By analyzing altruistic and biased planners alike, in regimes where their goals align with and where they oppose those of the agents, we span the full spectrum of these interactions rather than a single point within it.

Contributions and Outline This paper makes three principal contributions, each anchored in a concrete body of theoretical or empirical evidence.

1. **A theoretical framework for controlled social learning.** We introduce the first formal model to unify a dynamic control problem for a centralized information planner with the mechanism of sequential social learning. The planner strategically selects the precision of each agent’s private signal in pursuit of an objective that is either altruistic (maximizing social welfare) or biased (inducing a specified action), yielding a single framework that subsumes both benevolent and adversarial mediation within one dynamic program.

2. **A complete characterization of optimal planner policies.** We derive the optimal policy of both the altruistic and the biased planner as an explicit function of the evolving public belief. For the altruistic planner the characterization rests on a new proof that the value function is convex in the public belief—a result that is technically demanding precisely because the agents’ actions themselves depend on that belief, so that the standard linearity of expected reward in the belief state fails. The resulting policies expose, in closed form, the strategic trade-offs a planner confronts when its choices impose informational externalities on all future agents, and they establish that the biased planner optimally obfuscates information over an identifiable range of beliefs.
3. **Empirical study of the framework with LLMs.** We instantiate the model in controlled simulations—within a single car-purchasing scenario, using one open-weight LLM family for planner, agents, and oracle—and report three findings. (a) A planner that anticipates social learning shifts public opinion and social welfare far more than a myopic one, moving welfare by 40–50% relative to the no-control baseline in this setting. (b) The strategy the LLM planner adopts is structurally close to our non-trivial analytical optima, deviating by less than 10% across the majority of belief states even though the LLM agents update non-Bayesianly, indicating that the characterization is robust to the departures from rationality we observe. (c) The LLM planner’s systematic deviations from the optima track the specific, human-like cognitive biases of the agents it faces, consistent with the planner having acquired an implicit behavioral model of those agents. These findings, developed within one scenario and model family, are best read as an empirical demonstration of the framework’s descriptive reach rather than as validation across deployment settings; we delimit their scope explicitly in Section 6.

We situate the work relative to prior literature in Section 2, introduce the formal model in Section 3, and derive the optimal altruistic and biased policies in Sections 4 and 5. Section 6 evaluates these predictions through LLM-based simulations, and Section 7 concludes. Full proofs and additional results are provided in the appendices.

2 RELATED WORK

Our work draws together, and is distinguished from, four literatures: social learning, information design, online persuasion and reinforcement learning, and the strategic behavior of LLMs. In each case the point of departure is the same: prior work treats the sender’s problem and the observational-learning problem separately, whereas we couple them within a single dynamic program.

Social Learning Bikhchandani et al. (1992) and Banerjee (1992) introduced the sequential social learning framework in economics, and a substantial body of subsequent work has characterized its dynamics in progressively more general settings (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)). This literature has examined a wide range of structures for the social information available to agents, together with the biases and misperceptions to which agents are subject (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010), and Bikhchandani et al. (2024) and the references therein). Our model inherits this well-studied observational-learning substrate and augments it with an active, resource-constrained controller.

Work on the *control* of social learning, by contrast, remains comparatively sparse (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021), and the two studies closest to ours—Wei & Anastasopoulos (2022) and Smith et al. (2021)—differ from it in decisive respects. Wei & Anastasopoulos (2022) presupposes two-way communication, in which agents report their private signals to the planner in exchange for action recommendations, whereas Smith et al. (2021) treats a benevolent planner who directly overrides agents’ choice rules and derives a decentralized pivot mechanism to improve learning. Our formulation dispenses with two-way communication and leaves agents in full control of their own actions, a specification markedly better matched to modern information-mediating algorithms, which are typically invisible to, or black-boxed from, the users they influence.

Information Design In our setting the planner optimizes its objective by selecting the precision of each agent’s private signal, a constrained instance of the broader information-design problem (Bergemann & Morris, 2019). Such models rose to prominence with the Bayesian-persuasion framework of Kamenica & Gentzkow (2011), in which a sender commits to the signal structure that in-

forms a rational Bayesian agent about an unknown state. Our planner solves, in each period, a costly and constrained information-design problem: it selects a binary, symmetric signal and pays for any deviation from a baseline precision. Against the many existing variants of information design, the defining feature of our model is the explicit incorporation of social learning among the agents.

Only a handful of studies address information design within sequential social learning (Arieli et al., 2022; Wu et al., 2025), and both confine attention to a one-shot setting in which the sender commits at the outset to a single information structure that is then held fixed across the entire sequence of agents. Arieli et al. (2022) analyzes a self-interested sender who selects a general information structure in the manner of classic Bayesian persuasion, whereas Wu et al. (2025) constrains the structure to be Blackwell-more-informative and maximizes the probability of a correct cascade. We depart from both by studying a genuinely *dynamic* problem in which the sender chooses a fresh information structure for every agent in the sequence, thereby inducing a substantially richer class of sender strategies.

Online Persuasion and RL A rapidly growing literature sits at the interface of online learning and Bayesian persuasion. One strand studies online sequential persuasion under unknown environment parameters, in which a sender interacts repeatedly with short-lived receivers while ignorant of the prior over states and/or the receivers’ utilities (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al. (2023)). A second strand analyzes sequential persuasion in dynamic environments or MDPs, endowing the receiver with state transitions and/or foresight and emphasizing history-dependent or promise-based signaling policies with explicit connections to reinforcement learning (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). The unifying concern of these works is learning-based performance—sublinear regret against the best signaling scheme in hindsight, or approximation guarantees under computational constraints.

Our model differs categorically along three axes. First, the planner need not learn any unknown parameter: the sequential dependence originates entirely in social learning among the receivers, not in the sender updating beliefs about the environment. Second, the planner enjoys no informational advantage—it observes neither the underlying state nor the agents’ private signals—and yet, within this deliberately austere setting, we obtain a full characterization of its optimal policies. Third, although our problem is sequential, it is neither an online-learning nor a reinforcement-learning problem: actions do not alter the payoff-relevant state (there are no reward externalities), and the planner refines its knowledge solely by observing agents’ actions, which propagate belief through social learning rather than through payoff exploration or state evolution. The sequential coupling in our model is thus purely informational, arising because each agent’s action reshapes the beliefs of those who follow.

LLMs A recent line of work studies LLMs specifically within information design (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). These studies uniformly consider one-on-one interactions and therefore do not capture the social dynamics that are the object of our investigation. The closest to our own is Duetting et al. (2025), who study information design in which a sender chooses both a signal structure and a ‘framing’ that conditions the receiver’s—possibly non-Bayesian—posterior, and who introduce an LLM-based oracle to solve the computationally difficult problem of optimizing over the framing space. Their methodology rests on the premise, itself contested, that LLMs can emulate humans as economic agents (Dillion et al., 2023; Horton, 2023). Because their setting addresses neither the repeated interactions nor the social learning central to ours, it cannot speak to the coupled dynamics we study; we do, however, adapt components of their oracle-based methodology for the empirical study in Section 6.

3 SOCIAL LEARNING MODEL

We consider a planner facing a countable sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature draws a fixed, *unknown* exogenous state of the world $\omega \in \Omega := \{G, B\}$ with $\mathbb{P}(\omega = G) = b_1$, where the prior b_1 is common knowledge. In the recommendation-system example of Section 1, the state G denotes a good restaurant.

At each time i , the planner first determines how much to invest in personalization for agent i and then delivers to that agent a private signal $s_i \in \Omega$ to inform its decision. The signal is binary and matches the true state with a controllable probability $q_i \in [0.5, 1]$; that is, $\mathbb{P}(s_i = \omega) = q_i$. We

call q_i the *signal precision* of agent i ; it is the sole instrument through which the planner exercises control.

Conditional on ω , each agent's signal is independent of the signals of all other agents and of the history. On the basis of this signal together with the observed actions of the predecessors $j < i$, agent i selects an action $a_i \in \Omega$, receiving utility 0 when the action matches the true state ($a_i = \omega$) and $-C$ otherwise, with $C > 0$. Each agent acts to maximize her own expected utility, which in the running example amounts to patronizing the business precisely when the state is G .

3.1 AGENTS' DECISION PROBLEMS

Each agent observes the actions of all her predecessors together with their respective signal precisions, a history we denote $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Because the planner and every agent share the identical information set $\mathcal{H}_i$, they hold a common *public* belief about ω that is refined after each agent acts. The public belief immediately before agent i acts is $b_i := \mathbb{P}(\omega = G | \mathcal{H}_i)$, which coincides with the prior b_1 before any agent has acted. As in the classic models of Banerjee (1992) and Bikhchandani et al. (1992), the belief sequence $(b_i)_{i \in \mathbb{N}}$ is a Markov process and constitutes a sufficient statistic for the full history $\mathcal{H}_i$ (proof in Appendix B.3). This reduction is precisely what renders the planner's control problem tractable, casting it as a Markov decision process on the one-dimensional belief state.

Informed by $\mathcal{H}_i$ —equivalently, by the Markov property, by b_i —and by q_i , agent i selects the utility-maximizing action $a_i \in \Omega$, breaking ties in favor of the action matching s_i . She first forms a *private* belief $\tilde{b}_i$ about ω by combining $\mathcal{H}_i$ with her private signal s_i of precision q_i , a Bayesian update derived in Appendix B.1:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i} b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (1)$$

She then selects the action associated with the more probable state under this posterior: $a_i = G$ when $\tilde{b}_i > 0.5$, $a_i = B$ when $\tilde{b}_i < 0.5$, and $a_i = s_i$ at the tie $\tilde{b}_i = 0.5$. Substituting the expression for $\tilde{b}_i$ collapses this rule to a threshold in the public belief:

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Agent i 's action is subsequently observed by all later agents and folded into the updated public belief b_{i+1} according to (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. When $1 - q_i \leq b_i \leq q_i$, Equation (2) shows that agent i 's action perfectly reveals her private signal, so the updated public belief coincides exactly with her private belief. Outside this interval the signal is too weak to overturn the prevailing public belief: agent i 's action becomes uninformative, the public belief is frozen, and the process enters an absorbing state—the information cascade, or herding, of the classic social-learning literature (Bikhchandani et al., 1992; Banerjee, 1992). Once such a state is reached, society extracts no further private information from agents' actions, and learning halts permanently.

Agent i 's expected utility, $-C \mathbb{P}(a_i \neq \omega | b_i, q_i)$, admits the closed form (derived in Appendix C.1):

$$-C \mathbb{P}(a_i \neq \omega | b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

We analyze two archetypal planners: (1) an *altruistic* planner, which seeks to induce the correct action $a_i = \omega$ from every agent, and (2) a *biased* planner, which seeks to induce one fixed action, say G , regardless of ω . We index these planners by the subscripts A and B , respectively. In both

cases the planner’s only decision is the precision of each agent’s private signal, and the cost of that decision is captured by a function $\beta(\cdot)$ that is non-negative, increasing, continuous, and concave in its argument—so that informativeness is progressively more expensive to buy at the margin. Crucially, the planner is granted no informational advantage whatsoever: its information set is exactly the agents’ shared history $\mathcal{H}_i$.

Altruistic Planner At time i , the altruistic planner chooses precision q_i for agent i and incurs cost $\beta(q_i)$, normalized so that $\beta(p) = 0$ at a baseline precision $p \in [0.5, 1]$. Because the altruistic planner never benefits from reducing precision below this baseline (Appendix C.9), we may without loss restrict attention to $q_i \geq p$: cost is incurred only when precision is raised above p , and it then rises with further increases while exhibiting decreasing marginal cost. Both the baseline p and the cost function $\beta(\cdot)$ are common knowledge among the agents.

The altruistic planner maximizes social welfare—the expected total utility of the agents—net of the cost of its precision investments. Let $r_A(b_i, q_i)$ denote its instantaneous reward when it begins at public belief b_i and assigns precision q_i to agent i . Starting from prior b_1 and following a policy sequence $\pi = (\pi_i)_{i=1}^\infty$ with $\pi_i(\mathcal{H}_i) = q_i$, the planner attains, for a discount factor $\delta \in [0, 1]$, the expected total discounted utility

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C \mathbb{P}(a_i \neq \omega | b_i, q_i). \quad (5)$$

The altruistic planner’s optimal utility and optimal policy are, respectively, the supremum and arg-supremum of $V_A^\pi(\cdot)$ over the policy class Π . Setting $\delta = 0$ recovers the *myopic* value and policy as the special case in which the planner disregards all future externalities and optimizes each period in isolation.

Biased Planner The biased planner’s problem differs from the altruistic planner’s solely in its objective, and hence in its cost and reward functions; Appendix A develops a political-campaign instance that illustrates this planner concretely. The biased planner seeks action G from every agent irrespective of ω : it incurs cost 0 when an agent chooses G and cost $C > 0$ when the agent chooses B , in either case independently of the true state. What distinguishes this planner is that it may drive precision either *above* or *below* the baseline p , and both directions are costly. Any deviation from p requires tailoring the message to the individual agent—which in turn requires research into how that agent best assimilates content—so the biased planner incurs cost $\beta(|q_i - p|)$ for precision q_i , with $\beta(0) = 0$.

The biased planner’s expected instantaneous reward at time i is therefore

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = B | b_i, q_i) \quad (6)$$

The quantities $V_B^\pi(\cdot)$, $V_B^*(\cdot)$, and $\pi_B^*(\cdot)$ are then defined for the biased planner from Equation (6) exactly as $V_A^\pi(\cdot)$, $V_A^*(\cdot)$, and $\pi_A^*(\cdot)$ were defined for the altruistic planner from Equation (5).

The two objectives stand in a precise relationship to agent welfare. As established at the opening of Section 3, every agent is strictly better off choosing the action matching ω . Consequently, when $\omega = B$, the biased planner’s success *lowers* the agent’s utility, whereas the altruistic planner—which targets $a_i = \omega$ whatever the state—always *raises* it. The altruistic planner’s objective is thus aligned with each agent’s in every state, while the biased planner’s is aligned only when $\omega = G$. This single asymmetry is the source of every welfare effect we characterize.

Both planners’ problems are infinite-horizon discounted stationary Markov decision processes (MDPs) with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and transition kernel given by Equation (3) (Puterman, 1990), and each therefore admits a unique optimal value function (Kallenberg, 2011). We confine attention throughout to deterministic Markov policies, taking $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ with $\pi(b_i) = q_i$; the sufficiency of the public belief established above makes this restriction without loss for our purposes.

Remark 2. *Our model rests on three assumptions, each imposed deliberately to make the planner’s influence as constrained—and therefore its demonstrated power as conservative—as possible. Each is a genuine restriction in some settings and innocuous in others, and we state both boundaries explicitly: (1) The planner shares the agents’ history. This binds where the planner commands*

richer data, but is innocuous whenever all parties draw on the same public signals, such as early findings or survey data. (2) The planner may alter agents' signal precision, yet signals still pass through a binary symmetric channel. This binds where planners can cherry-pick, censor, or fabricate content, but is innocuous where their role is confined to improving the quality of noisy sources without altering their content. (3) The planner's control choices are fully observable. This binds where covert policies or framing are feasible, but is innocuous in transparent environments. Because these restrictions circumscribe the planner far more tightly than a real mediator would be, the substantial welfare effects we obtain under all three should be read as a lower bound on the influence such a mediator could exert—which is precisely what makes the findings striking.

4 OPTIMAL ALTRUISTIC POLICIES

We begin with the myopic case, which isolates the planner's problem from social learning by setting the discount factor to $\delta = 0$, so that all future externalities are ignored and $V_A^\pi(b) = r_A(b, \pi(b))$. This benchmark is the foil against which we later measure the value of anticipating social learning. The optimal myopic policy $\pi_A^0(\cdot)$ is

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

The myopic altruistic problem admits an equivalent decentralized interpretation: it is exactly the problem in which each agent chooses the precision of her own private signal, bearing the associated cost, so as to maximize her own expected utility net of that cost. The myopic planner, in other words, internalizes no externality on others, which is what makes it the natural baseline for isolating the value of foresight.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \quad \text{where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

The myopic optimum is therefore a clean threshold policy: when the public belief is sufficiently strong the planner defaults to the free baseline precision p , and when the belief is weak it supplies a perfect signal of precision 1. The threshold t_M depends only on the cost of the perfect signal, $\beta(1)$, relative to the cost of an incorrect action, C . When $\beta(1) \geq C$ the perfect signal is prohibitively expensive and is never used; $t_M = 0.5$, and the belief interval on which precision 1 is myopically optimal collapses to the empty set.

The non-myopic optimum demands a substantially more delicate analysis, which we anchor in the following structural property of the value function.

Theorem 2. *$V_A^*(\cdot)$ is convex with respect to public belief.*

The proof of Theorem 2 (Appendix C.3) is intricate and, we believe, of independent interest. Its difficulty stems directly from the dependence of the agents' actions on the public belief. Were actions independent of the belief process—as in Nyarko (1994)—the expected utility would be linear in the belief state and convexity of the value function would follow immediately. In our setting no such shortcut exists, and we instead establish convexity inductively, invoking only the martingale property of Bayesian updates and the convexity of the instantaneous reward. This convexity is the linchpin of our policy characterization.

Theorem 3. *There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and*

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A) \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & \text{o.w.} \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

The optimal policy partitions the belief space into three interpretable phases. In the first, as under the myopic optimum, the planner declines to invest at extreme values of the public belief; tellingly,

however, the non-myopic optimum demands a *stronger* public belief before disinvesting ($d_A < t_M$ whenever $t_M < 0.5$), precisely because—unlike its myopic counterpart—it internalizes the value of the information its investment would create for future agents.

In the second phase, when the public belief lies near 0.5 and thus carries little information, the planner selects the perfect precision 1 whenever it is not cost-prohibitive. By Equation (2) the agent’s action then matches the true state with probability 1, and the public belief collapses decisively to 0 or 1—resolving the collective uncertainty in a single step.

In the third phase, the planner chooses the minimal precision $\max(b, 1 - b)$ at which the agent’s action still reflects her private signal (see Equation (2)). Any lower precision would render the action uninformative—conveying nothing beyond what peers already know—so this is the cheapest precision that keeps social learning alive through the observation of peers’ actions.

5 OPTIMAL BIASED POLICIES

As in the altruistic analysis of Section 4, we begin with the myopic optimum. The myopic biased policy $\pi_B^0(\cdot)$ is again the arg-supremum of the instantaneous reward $r_B(b, q)$, but its structure proves considerably richer than in the altruistic case. Characterizing $\pi_B^0(\cdot)$ requires confronting an essential subtlety absent from the altruistic case: an exactly optimal policy need not exist in the biased setting (Puterman, 1990, sec. 2.3.1), because the state-action value can be discontinuous in the control. Where an optimum fails to exist we instead characterize ϵ -optimal policies, denoted by an ϵ superscript.

Theorem 4. *There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:*

- (A) *If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.*
- (B) *If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.*
- (C) *If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.*
- (D) *If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Proof in Appendix C.6. We now apply Theorem 4 to characterize the optimal biased policy.

Theorem 5. *There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:*

- (A) *If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.*
- (B) *If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.*
- (C) *If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$.*
- (D) *If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.*
- (E) *If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon, *}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Theorem 5 (proven in Appendix C.8 and depicted in figure 2a) reveals a rich five-phase strategy in which the biased planner alternates between amplifying and deliberately degrading information according to the prevailing belief. We read the phases in order of increasing belief.

When the public belief is already extreme (phase (A)), steering the system would cost more than it is worth, even given the adverse consequences for the planner’s payoff when $b < 0.5$. The chosen precision p then lies below $\max(b, 1 - b)$, so by Equation (2) the agent conforms to the public belief regardless of her private signal. That action is uninformative and, the policy being Markovian, leaves the belief unchanged in perpetuity—an unfavorable cascade when $b < 0.5$ and a favorable one otherwise.

When the belief approaches an unfavorable cascade (phase (B)), the planner raises precision high enough to move the agent’s action even though the realized signal is expected to be B . The agent then follows her private signal, which retains a non-zero probability of favoring the planner—a calculated last-ditch effort to escape the impending unfavorable cascade.

For slightly higher beliefs (phase (C)), the planner instead *lowers* precision below p . Because $b < 0.5$ here, more precise signals more often deliver unfavorable news; the planner therefore holds precision just high enough to influence the action ($q \geq 1 - b$), deploying the least informative such signal in the hope of drifting toward a more favorable belief.

When the belief weakly favors the planner’s target action (phase (D)), the planner invests at precision at least p : with $b > 0.5$, greater precision makes the agent more likely to infer $\omega = G$, so investment here reinforces an already favorable belief.

At still higher beliefs (phase (E)), the planner shades precision just below $\max(b, 1 - b)$ to $b - \epsilon$, so that agents disregard their private signals and take action G . Here the danger that a private signal overturns the favorable public belief outweighs both the cost of reducing precision and the marginal gain from pushing the belief still higher. This phase is the biased planner’s deliberate obfuscation of information at high beliefs—the theoretical counterpart of the welfare losses we document empirically in Section 6.

6 EVALUATION VIA LLM-BASED SIMULATION

We now deploy LLMs to probe, empirically, both the implications and the descriptive reach of our model and its theoretical predictions. The evaluation proceeds in three stages: we first characterize the belief-updating behavior of LLM agents to identify their systematic departures from Bayesian rationality, then examine how an LLM planner adapts its strategy to those departures, and finally quantify the consequences for social welfare. We are explicit at the outset about the scope of this evidence: all results below are obtained within a single car-purchasing scenario and a single open-weight LLM family, so they should be read as a controlled demonstration that the framework’s structural predictions survive genuinely non-Bayesian behavior, not as a claim of external validity across tasks, populations, or model families. Section 7 returns to this boundary.

To realize the controlled-social-learning setting concretely, we instantiate the model in a scenario in which a sequence of agents decides whether to buy a new car. As shown in figure 1a, LLMs occupy three distinct roles, each described below with the corresponding prompts given in Appendix E.2.2:

Agent: role-playing an assigned profile, each agent observes the actions of its predecessors together with a private message of known precision about the car, updates its belief about the car’s quality, and decides whether to buy.

Planner: having observed the history of actions, the planner selects the precision q_i of agent i ’s private signal in accordance with its objective.

Oracle: given the planner’s chosen precision, an agent profile, and a fact sheet about the car (see Appendix E.2.1), the oracle produces a private signal of the requested precision, tailored to that agent. This role factorization lets us hold the information structure of Section 3 fixed while varying only the component under study. We compare the outcomes of these LLM-based simulations against exact numerical solutions of the planner’s MDP. In both the LLM and the numerical experiments we adopt a linear cost function, $\beta(q) = k|q - p|$, and sweep the cost coefficient k , the baseline precision p , and the discount factor δ , holding the cost of an incorrect action fixed at $C = 1$; the full parameter grid is given in Table 1. Appendix E details the experimental setup and prompting, and Appendix E.3 independently validates both the agents’ self-reported beliefs and the oracle’s precision control.

Within this single scenario and model family, the experiments support three findings, stated here with the scope each is entitled to:

1. **Anticipating social learning is decisive.** Both the analytical and the LLM planner substantially improve their own utility by anticipating and exploiting social-learning dynamics, and this leverage translates into large swings in social welfare—beneficial or harmful according to whether the planner’s and the agents’ objectives are aligned.
2. **The analytical characterization is robust to non-Bayesian agents.** LLM planners adopt policies strikingly close to our non-obvious analytical optima even while facing genuinely non-Bayesian LLM agents, indicating that the theoretical characterization survives the departures from the rationality assumptions under which it was derived.

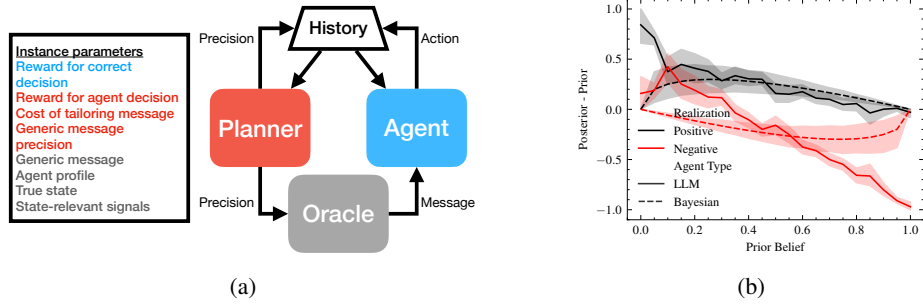


Figure 1: (a) System diagram for our experiments. The three LLM roles—planner, agent, and oracle—share the public history; instance parameters are color-coded by the role that consumes them. (b) Post-signal belief revision for LLM (solid) and Bayesian (dashed) agents at a fixed prior, following a positive (black) or negative (red) signal. The systematic gap between the LLM and Bayesian curves quantifies the LLM agents’ departure from Bayesian updating and sets up the non-Bayesian patterns analyzed in Section 6.1.

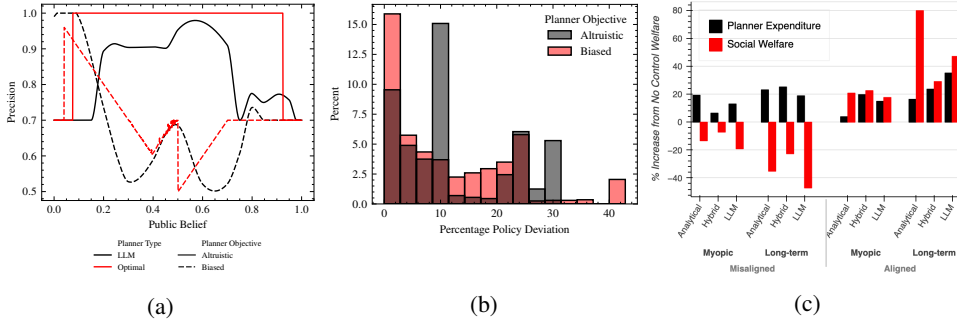


Figure 2: (a) Representative policies of the LLM planner (black) and the analytically optimal planner (red) in the altruistic (solid) and biased (dashed) settings. The two agree closely in structure even for the non-trivial, multi-phase biased policy. (b) Distribution, across the public belief space, of the percentage deviation between the LLM and optimal policies for both objectives; the mass concentrated below 10% quantifies the alignment visible in (a). (c) Planner expenditure and change in social welfare, each expressed as a percentage of the no-control baseline welfare, with the true state fixed to B . The left half reports a biased planner seeking action G and the right an altruistic planner; the two halves move welfare sharply in opposite directions, isolating the effect of the planner’s objective while holding the mediation mechanism fixed.

3. The LLM planner’s deviations track agent biases. Beyond tracking the optimal policies, LLM planners deviate from them in systematic, interpretable ways that compensate for the specific non-Bayesian behavior of the agents they face—a pattern consistent with the planner having acquired, without explicit instruction, an implicit behavioral model of those agents.

6.1 LLMs AS NON-BAYESIAN AGENTS

Although our analytical results assume Bayesian agents, both humans and LLMs are well documented to exhibit systematic cognitive biases (Rupprecht et al., 2025). To isolate the biases specific to our LLM agents, we measure how an agent’s stated belief responds to new information: we supply a fixed prior together with a private signal of known precision and elicit the resulting posterior. These measurements, reported in figure 1b, expose three consistent non-Bayesian patterns in the LLM agents’ belief updating (corroborated at the trajectory level in Appendix E.4):

- (NB1) LLM agents underreact to private signals which align with their prior beliefs.
- (NB2) LLM agents overreact to private signals which run counter to their prior beliefs.

(NB3) As a result, LLM agents require a stronger public belief to enter an information cascade.

These same biases have been documented empirically in human subjects (e.g., Ba et al. (2022); Chan et al. (2025)), which indicates that our LLM agents reproduce human-like, rather than idiosyncratic, departures from rationality—though we make this comparison at the level of qualitative pattern, not calibrated magnitude. The following sections show that the LLM planner’s policy remains close to the analytical optimum in the face of non-Bayesian agents of exactly this kind.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

We now compare the policies chosen by LLM planners against the optima derived in Sections 4 and 5. As figure 2a shows, and notwithstanding the agents’ non-Bayesian behavior, the LLM strategies exhibit a close structural correspondence to the theoretical optimum. Even where that optimum is highly non-trivial—as in the multi-phase biased policy of figure 2a—the LLM planner reproduces its qualitative structure, including the belief-dependent alternation between amplifying and degrading information, which suggests that the planner is responding to the social-learning dynamics of the sequence rather than to individual agents in isolation.

The correspondence holds across both objectives. In the altruistic case, both planners withhold investment once the public belief is strong and invest heavily when it is uncertain. In the biased case, both trace the same qualitative arc: aggressive investment to escape an unfavorable belief, reduced precision near the midpoint, and cessation of investment once a favorable belief is secured. The agreement is quantitative as well as qualitative: as figure 2b shows, the percentage deviation between the LLM and optimal policies falls below 10% across the majority of belief states for both planners, corroborating the structural alignment visible in figure 2a.

The residual differences are themselves informative and are best read as the planner’s adaptations to the specific non-Bayesian behaviors of Section 6.1. (1) The LLM planner avoids the extreme precisions 0.5 and 1.0, consistent with a documented central-tendency bias (Rupprecht et al., 2025). (2) Its investment tapers gradually rather than terminating at a sharp threshold—a response to the agents’ resistance to cascades (NB3), since it is never fully “safe” to stop investing when even strongly primed agents can be swayed by a countervailing private signal. (3) The biased LLM planner keeps investing even at very low beliefs, consistent with an implicit expectation that its agents may overreact (NB2) to a surprisingly positive signal, which renders a last-ditch effort more viable than it would be under Bayesian agents. Each deviation thus maps onto a specific documented bias, a correspondence consistent with the planner having, in effect, fitted a behavioral model of the agents it seeks to influence.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

We define social welfare $W^\pi(b)$ as the total expected discounted utility of all agents. Because more informative signals never lower an agent’s expected utility, welfare is monotone in signal precision (Appendix C.9); it follows immediately that the altruistic planner always raises welfare above the no-control baseline. To disentangle the contribution of the policy from that of the agent model, we compare social welfare and planner expenditure across three settings that vary these two factors independently: the *analytical* setting (optimal policy, Bayesian agents), the *LLM* setting (LLM planner, LLM agents), and a *hybrid* setting (optimal policy, LLM agents).

The results in figure 2c show that in every setting the planner moves social welfare substantially, and that neglecting social learning—as the myopic policies do—markedly degrades the planner’s outcomes. Most striking is the misaligned case: both the analytical and the LLM biased planner depress social welfare by 40 to 50% relative to the no-control baseline, achieved by deliberately obscuring information about the true state through the precision-reducing policies of phases (C) and (E) in Theorem 5. That harm of this magnitude is attainable at all is the more notable because these planners operate under the stringent constraints of Remark 2—information parity, no fabrication, and full observability—so the figure gives a concrete lower bound on the risk a misaligned LLM information mediator would pose in this scenario.

The comparison also exposes the cost of acting on a misspecified model of the agents. In the hybrid setting the analytically optimal policy, derived for Bayes-rational agents, proves brittle: its welfare degrades when applied to the non-Bayesian LLM agents. The LLM policy, by contrast, retains the

structure of the analytical policy while adjusting to agents with human-like biases, and is correspondingly better matched to this more realistic population.

7 CONCLUSION

We introduced a framework for the strategic control of social learning by an algorithmic information mediator and developed a formal model of this increasingly consequential dynamic. We proved that even a tightly constrained planner can decisively shift social welfare in either direction, and that anticipating social learning is integral to exercising this influence effectively. Controlled LLM simulations corroborated these predictions within a single scenario, showing that an LLM planner adopts policies reproducing the structure of our analytical optima while adapting to non-Bayesian agents.

Taken together, these results yield three takeaways for the study of algorithmic information mediation. First, the impact of human–AI or human–algorithm interaction cannot be assessed in isolation: neglecting its interplay with concurrent dynamics such as social learning systematically understates its potential, whether beneficial or harmful. Second, information mediators can guide or derail social learning and public opinion even when their influence is subtle and constrained, which makes the design of mitigating safeguards a priority rather than an afterthought. Third, the LLM planner in our simulations exhibits strategic behavior that both accounts for and capitalizes on social learning and on non-Bayesian cognitive biases. The latter two points sharpen the need to understand how the risks of algorithmic mediators—and of LLMs in particular—might be contained.

These same properties carry a direct dual-use hazard that we wish to name explicitly. The biased-planner analysis and the obfuscation policy it prescribes are, read adversarially, a recipe: the very characterization that lets a regulator anticipate a manipulative mediator could equally guide one, whether in political persuasion, commercial steering, or the selective withholding and personalization of information. We regard the appropriate response as asymmetric—the framework’s value lies in making such strategies legible to those who would detect and constrain them—and we deliberately restrict the empirical study to a benign car-purchasing scenario with synthetic profiles, releasing no capability that is not already implicit in deployed recommendation and persuasion systems. We nonetheless caution that the results should not be operationalized to optimize manipulative mediation, and that the design of countervailing safeguards should proceed in step with any such work.

Two limitations bound the strength of our empirical claims and should temper their interpretation. First, the simulations use a single task, a single open-weight LLM family, and self-reported beliefs; sensitivity to task, prompt phrasing, decoding temperature, and model family remains to be mapped, and our claims of human-likeness are qualitative rather than calibrated. Second, and relatedly, the study contains no human data; because the fidelity of LLM-based simulators of human behavior remains contested (see, e.g., Horton (2023); Gao et al. (2025)), human data would be valuable both for validating and potentially for fine-tuning these simulators. We view a systematic evaluation across scenarios, model families, and decoding settings, together with comparison against non-LLM algorithmic planners, as the natural next step for establishing the external validity that the present controlled study does not claim.

We close by highlighting two directions we regard as especially promising: generalizing the model and its results, and mitigating the adverse welfare effects of biased planners. Several generalizations would relax our modeling assumptions and bring the framework closer to reality—larger state spaces, more general signal structures, more general loss functions, and heterogeneous agents. Moving beyond binary states and 0-1 losses poses difficulties that are primarily algebraic, and we conjecture that the qualitative structure of our results persists in those richer settings. For more general signal structures the central obstacle is preserving the convexity result of Theorem 2; identifying the broadest class of signal structures for which convexity continues to hold is, in our view, a particularly compelling problem, since convexity is what underwrites the entire policy characterization. As a first concrete step toward heterogeneity, we already extend the altruistic-planner results to populations containing agents with opposing preferences in Appendix D.

A second direction, complementary to the first, is the design of interventions that forestall the welfare losses a biased planner inflicts. A natural avenue is to devise regulations or mechanisms that realign the incentives of planners and agents so as to avert detrimental welfare effects; the strategic

characterization we provide here is a necessary first step toward such safeguards, in that one cannot regulate a strategy one cannot yet describe.

REFERENCES

- Shipra Agrawal, Yiding Feng, and Wei Tang. Dynamic Pricing and Learning with Bayesian Persuasion. *Advances in Neural Information Processing Systems*, 36:59273–59285, December 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/b9c2e8a0bbbed5fcfaf62856a3a719ada-Abstract-Conference.html.
- Lisa P. Argyle, Christopher A. Bail, Ethan C. Busby, Joshua R. Gubler, Thomas Howe, Christopher Rytting, Taylor Sorensen, and David Wingate. Leveraging AI for democratic discourse: Chat interventions can improve online political conversations at scale. *Proceedings of the National Academy of Sciences*, 120(41):e2311627120, October 2023. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.2311627120. URL <https://pnas.org/doi/10.1073/pnas.2311627120>.
- Itai Arieli and Manuel Mueller-Frank. A general analysis of sequential social learning. *Mathematics of Operations Research*, 46:1235–1249, 11 2021.
- Itai Arieli, Ronen Gradwohl, and Rann Smorodinsky. Herd Design. In *Proceedings of the 23rd ACM Conference on Economics and Computation*, pp. 871–872, Boulder CO USA, July 2022. ACM. ISBN 978-1-4503-9150-4. doi: 10.1145/3490486.3538292. URL <https://dl.acm.org/doi/10.1145/3490486.3538292>.
- Cuimin Ba, J. Aislinn Bohren, and Alex Imas. Over- and Underreaction to Information. *SSRN Electronic Journal*, 2022. ISSN 1556-5068. doi: 10.2139/ssrn.4274617. URL <https://www.ssrn.com/abstract=4274617>. Publisher: Elsevier BV.
- Francesco Bacchiocchi, Matteo Bollini, Matteo Castiglioni, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persuasion Without a Clue. *Advances in Neural Information Processing Systems*, 37:76404–76449, December 2024.
- Fengshuo Bai, Mingzhi Wang, Zhaowei Zhang, Boyuan Chen, Yinda Xu, Ying Wen, and Yaodong Yang. Efficient Model-agnostic Alignment via Bayesian Persuasion, May 2024.
- Abhijit V Banerjee. A simple model of herd behavior. *The Quarterly Journal of Economics*, 107(3):797–817, 1992.
- Larry M. Bartels. *Presidential primaries and the dynamics of public choice*. Princeton, N.J. : Princeton University Press, 1988.
- Chris T Bauch and Samit Bhattacharyya. Evolutionary game theory and social learning can determine how vaccine scares unfold. *PLoS computational biology*, 8(4), 2012.
- Dirk Bergemann and Stephen Morris. Information Design: A Unified Perspective. *Journal of Economic Literature*, 57(1):44–95, March 2019. ISSN 0022-0515. doi: 10.1257/jel.20181489. URL <https://www.aeaweb.org/articles?id=10.1257/jel.20181489>.
- Martino Bernasconi, Matteo Castiglioni, Alberto Marchesi, and Mirco Mutti. Persuading Farsighted Receivers in MDPs: The Power of Honesty. *Advances in Neural Information Processing Systems*, 36:14987–15014, December 2023.
- Sujay Bhatt and Vikram Krishnamurthy. Controlled Sequential Information Fusion With Social Sensors. *IEEE Transactions on Automatic Control*, 66(12):5893–5908, December 2021.
- Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.
- Sushil Bikhchandani, David Hirshleifer, Omer Tamuz, and Ivo Welch. Information Cascades and Social Learning. *Journal of Economic Literature*, 62(3):1040–1093, September 2024. ISSN 0022-0515, 2328-8175. doi: 10.1257/jel.20241472. URL <https://pubs.aeaweb.org/doi/10.1257/jel.20241472>.
- Ilai Bistriz, Nasim Heydaribeni, and Achilleas Anastasopoulos. Informational cascades with nonmyopic agents. *IEEE Transactions on Automatic Control*, 67, 9 2022.
- William J Brady, Killian McLoughlin, Tuan N Doan, and Molly J Crockett. How social learning amplifies moral outrage expression in online social networks. *Science Advances*, 7(33), 2021.

- Carlos Carrasco-Farre. Large Language Models are as persuasive as humans, but how? About the cognitive effort and moral-emotional language of LLM arguments, April 2024. URL <http://arxiv.org/abs/2404.09329>. arXiv:2404.09329 [cs].
- Matteo Castiglioni, Andrea Celli, Alberto Marchesi, and Nicola Gatti. Online Bayesian Persuasion. In *Advances in Neural Information Processing Systems*, volume 33, pp. 16188–16198. Curran Associates, Inc., 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/ba5451d3c91a0f982f103cdbc249bc78-Abstract.html>.
- Matteo Castiglioni, Alberto Marchesi, Andrea Celli, and Nicola Gatti. Multi-Receiver Online Bayesian Persuasion. In *Proceedings of the 38th International Conference on Machine Learning*, pp. 1314–1323. PMLR, July 2021.
- Kenneth Chan, Gary Charness, Chetan Dave, and J. Lucas Reddinger. On Prior Confidence and Belief Updating, May 2025. URL <http://arxiv.org/abs/2412.10662>. arXiv:2412.10662 [econ].
- Zirui Cheng and Jiaxuan You. Towards Strategic Persuasion with Language Models, September 2025.
- Krishna Dasaratha and Kevin He. Learning from viral content. *arXiv preprint arXiv:2210.01267*, 2022.
- Danica Dillion, Niket Tandon, Yuling Gu, and Kurt Gray. Can AI language models replace human participants? *Trends in Cognitive Sciences*, 27(7):597–600, July 2023. ISSN 1364-6613, 1879-307X. doi: 10.1016/j.tics.2023.04.008. URL [https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613\(23\)00098-0](https://www.cell.com/trends/cognitive-sciences/abstract/S1364-6613(23)00098-0). Publisher: Elsevier.
- Paul Duetting, Safwan Hossain, Tao Lin, Renato Paes Leme, Sai Srivatsa Ravindranath, Haifeng Xu, and Song Zuo. Information Design With Large Language Models, September 2025. URL <http://arxiv.org/abs/2509.25565>. arXiv:2509.25565 [cs].
- Maxim Egorov, Zachary N. Sunberg, Edward Balaban, Tim A. Wheeler, Jayesh K. Gupta, and Mykel J. Kochenderfer. POMDPs.jl: A framework for sequential decision making under uncertainty. *Journal of Machine Learning Research*, 18(26):1–5, 2017.
- Erik Eyster and Matthew Rabin. Naïve herding in rich-information settings. *American Economic Journal: Microeconomics*, 2:221–243, 11 2010.
- Shen Gao, Jiabao Fang, Quan Tu, Zhitao Yao, Zhumin Chen, Pengjie Ren, and Zhaochun Ren. Generative News Recommendation. In *Proceedings of the ACM Web Conference 2024*, WWW ’24, pp. 3444–3453, New York, NY, USA, May 2024. Association for Computing Machinery. ISBN 979-8-4007-0171-9. doi: 10.1145/3589334.3645448. URL <https://dl.acm.org/doi/10.1145/3589334.3645448>.
- Yuan Gao, Dokyun Lee, Gordon Burtch, and Sina Fazelpour. Take Caution in Using LLMs as Human Surrogates: Scylla Ex Machina, January 2025. URL <http://arxiv.org/abs/2410.19599>. arXiv:2410.19599 [econ].
- Kenneth T Gillingham and Bryan Bollinger. Social learning and solar photovoltaic adoption. *Management Science*, 67(11):7091–7112, 2021.
- Douglas Guilbeault, Joshua Becker, and Damon Centola. Social learning and partisan bias in the interpretation of climate trends. *Proceedings of the National Academy of Sciences*, 115(39):9714–9719, 2018.
- Keegan Harris, Nicole Immorlica, Brendan Lucier, and Aleksandrs Slivkins. Algorithmic Persuasion Through Simulation, February 2025. URL <http://arxiv.org/abs/2311.18138>. arXiv:2311.18138 [cs].
- Claudia E. Haupt and Mason Marks. AI-Generated Medical Advice—GPT and Beyond. *JAMA*, 329(16): 1349–1350, April 2023. ISSN 0098-7484. doi: 10.1001/jama.2023.5321. URL <https://doi.org/10.1001/jama.2023.5321>.
- K. Hinderer. Lipschitz Continuity of Value Functions in Markovian Decision Processes. *Mathematical Methods of Operations Research*, 62(1):3–22, September 2005.
- John J. Horton. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?, April 2023. URL <https://www.nber.org/papers/w31122>.
- Bright Huo, Amy Boyle, Nana Marfo, Wimonchat Tangamornsuksan, Jeremy P. Steen, Tyler McKechnie, Yung Lee, Julio Mayol, Stavros A. Antoniou, Arun James Thirunavukarasu, Stephanie Sanger, Karim Ramji, and Gordon Guyatt. Large Language Models for Chatbot Health Advice Studies: A Systematic Review. *JAMA Network Open*, 8(2):e2457879, February 2025. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2024.57879. URL <https://doi.org/10.1001/jamanetworkopen.2024.57879>.

756 Cameron R. Jones and Benjamin K. Bergen. Lies, Damned Lies, and Distributional Language Statistics: Per-
757 suasion and Deception with Large Language Models, December 2024. URL [http://arxiv.org/abs/](http://arxiv.org/abs/2412.17128)
758 2412.17128. arXiv:2412.17128 [cs].

759 Lodewijk Kallenberg. Markov decision processes. *Lecture Notes. University of Leiden*, 428, 2011.

760

761 Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–
762 2615, 2011.

763 Vikram Krishnamurthy. Quickest Detection POMDPs With Social Learning: Interaction of Local and Global
764 Decision Makers. *IEEE Transactions on Information Theory*, 58(8):5563–5587, August 2012.

765

766 LG AI Research. Exaone deep: Reasoning enhanced language models. *arXiv preprint arXiv:2503.12524*, 2025.

767

768 Minqian Liu, Zhiyang Xu, Xinyi Zhang, Hejun An, Sarvech Qadir, Qi Zhang, Pamela J. Wisniewski, Jin-Hee
769 Cho, Sang Won Lee, Ruoxi Jia, and Lifu Huang. LLM Can be a Dangerous Persuader: Empirical Study
770 of Persuasion Safety in Large Language Models, April 2025. URL [http://arxiv.org/abs/2504.](http://arxiv.org/abs/2504.10430)
771 10430. arXiv:2504.10430 [cs].

772 Nunzio Lorè and Babak Heydari. Strategic behavior of large language models and the role of game struc-
773 ture versus contextual framing. *Scientific Reports*, 14(1), August 2024. ISSN 2045-2322. doi: 10.1038/
774 s41598-024-69032-z. URL <https://www.nature.com/articles/s41598-024-69032-z>.
775 Publisher: Springer Science and Business Media LLC.

776 Yaw Nyarko. On the convexity of the value function in Bayesian optimal control problems. *Economic Theory*,
777 4(2):303–309, March 1994.

778 Kenneth Payne and Baptiste Alloui-Cros. Strategic Intelligence in Large Language Models: Evidence from evo-
779 lutionary Game Theory, July 2025. URL <http://arxiv.org/abs/2507.02618>. arXiv:2507.02618
780 [cs].

781 Yujin Potter, Shiyang Lai, Junsol Kim, James Evans, and Dawn Song. Hidden Persuaders: LLMs’ Political
782 Leaning and Their Influence on Voters, November 2024.

783

784 Martin L. Puterman. Chapter 8 markov decision processes. In *Stochastic Models*, volume 2 of *Handbooks in*
785 *Operations Research and Management Science*. Elsevier, 1990.

786 Maya Raifer, Guy Rotman, Reut Apel, Moshe Tennenholtz, and Roi Reichart. Designing an Automatic Agent
787 for Repeated Language-based Persuasion Games. *Transactions of the Association for Computational Lin-*
788 *guistics*, 10:307–324, March 2022.

789 Neel Rao, Markus M Mobius, and Tanya Rosenblat. Social networks and vaccination decisions. *FRB of Boston*,
790 2007.

791

792 Alexander Rogiers, Sander Noels, Maarten Buyl, and Tijl De Bie. Persuasion with Large Language Models: a
793 Survey, November 2024. URL <http://arxiv.org/abs/2411.06837>. arXiv:2411.06837 [cs].

794 Jens Rupperecht, Georg Ahnert, and Markus Strohmaier. Prompt Perturbations Reveal Human-Like Biases in
795 LLM Survey Responses, July 2025. URL <http://arxiv.org/abs/2507.07188>. arXiv:2507.07188
796 [cs].

797 Dr.-Ing. Martin Schiele, Yannick Gittmann, Stefan Ilchmann, Ante Gojsalić, Dominik Ju-
798 rinčić, and Phylis Klempt. Voting Advice Applications: Implementation of RAG-supported
799 LLMs, July 2024. URL [https://www.techrxiv.org/users/802685/articles/](https://www.techrxiv.org/users/802685/articles/1187651-voting-advice-applications-implementation-of-rag-supported-llms?commit=c540a54749906f8ac761093f5dff0f514291d4b6)
800 1187651-voting-advice-applications-implementation-of-rag-supported-llms?
801 commit=c540a54749906f8ac761093f5dff0f514291d4b6.

802 James E. Smith and Kevin F. McCardle. Structural Properties of Stochastic Dynamic Programs. *Operations*
803 *Research*, 50(5):796–809, October 2002.

804

805 Lones Smith and Peter Sørensen. Pathological outcomes of observational learning. *Econometrica*, 68:371–398,
806 2000.

807

808 Lones Smith, Peter Norman Sørensen, and Jianrong Tian. Informational Herding, Optimal Experimentation,
809 and Contrarianism. *The Review of Economic Studies*, 88(5):2527–2554, September 2021. ISSN 0034-
6527, 1467-937X. doi: 10.1093/restud/rdab001. URL [https://academic.oup.com/restud/](https://academic.oup.com/restud/article/88/5/2527/6149316)
article/88/5/2527/6149316.

-
- Shih-Tang Su, Vijay G. Subramanian, and Grant Schoenebeck. Bayesian Persuasion in Sequential Trials. In Michal Feldman, Hu Fu, and Inbal Talgam-Cohen (eds.), *Web and Internet Economics*, pp. 22–40, Cham, 2022. Springer International Publishing.
- Jeremy G Weber. Social learning and technology adoption: The case of coffee pruning in peru. *Agricultural Economics*, 43:73–84, 2012.
- Xupeng Wei and Achilleas Anastasopoulos. Social learning with a self-interested coordinator. In *2022 IEEE 61st Conference on Decision and Control (CDC)*, pp. 3137–3144. IEEE, 2022.
- Marcus Williams, Micah Carroll, Adhyyan Narang, Constantin Weissner, Brendan Murphy, and Anca Dragan. On Targeted Manipulation and Deception when Optimizing LLMs for User Feedback, February 2025. URL <http://arxiv.org/abs/2411.02306>. arXiv:2411.02306 [cs].
- Jibang Wu, Zixuan Zhang, Zhe Feng, Zhaoran Wang, Zhuoran Yang, Michael I. Jordan, and Haifeng Xu. Sequential Information Design: Markov Persuasion Process and Its Efficient Reinforcement Learning, February 2022.
- Shuo Wu, Pawan Poojary, and Randall Berry. Observational Learning with a Budget, April 2025. URL <http://arxiv.org/abs/2504.19396>. arXiv:2504.19396 [cs].
- Tianyi Zhang, Faisal Ladhak, Esin Durmus, Percy Liang, Kathleen McKeown, and Tatsunori B. Hashimoto. Benchmarking Large Language Models for News Summarization. *Transactions of the Association for Computational Linguistics*, 12:39–57, January 2024a. ISSN 2307-387X. doi: 10.1162/tacl.a_00632. URL https://doi.org/10.1162/tacl.a_00632.
- Yadong Zhang, Shaoguang Mao, Tao Ge, Xun Wang, Adrian de Wynter, Yan Xia, Wenshan Wu, Ting Song, Man Lan, and Furu Wei. LLM as a Mastermind: A Survey of Strategic Reasoning with Large Language Models, April 2024b. URL <http://arxiv.org/abs/2404.01230>. arXiv:2404.01230 [cs].

A APPLICATIONS ELABORATED

Consider a sequence of individuals (agents, in our terminology) deciding whether to patronize a business or service provider—a restaurant, contractor, or realtor, say—together with a recommendation system (the planner) that serves them information about that business. The unknown binary state of the world is whether the business is good. Both parties wish to act consistently with this state: agents want to patronize the business precisely when it is good, and the recommendation system wants to induce them to do so. In this alignment of objectives the recommendation system is an *altruistic planner*.

The recommendation system may show an agent a highly targeted—*precise*, in our terminology—advertisement that presents the strengths and weaknesses of the business in contexts the agent can relate to given her background and characteristics, or it may show a generic, confusing advertisement that does little to inform her action. The precise signal is more informative, comprehensible, and relatable, and hence more likely to steer the agent to the correct decision—patronizing the business if and only if it is good—so social welfare rises as more agents decide correctly. Raising an advertisement’s precision is itself costly, since it requires tailoring to the agent’s specific background, so the recommendation system chooses the precisions that maximize social welfare net of these costs.

We now give a real-world instance of a *biased planner*. Consider a body of voters (agents) motivated to support whichever candidate is most likely to prevail in their community. This impulse to “back a winner” is a documented driver of electoral outcomes—indeed, it is why U.S. states holding early primaries exert an outsized effect on results (see Bartels (1988)). The planner here is a particular candidate’s campaign, which accordingly seeks to induce agents to back that candidate. The binary state of the world encodes whether the candidate is winning, so an agent who knew the true state would back the candidate if and only if he were winning.

Each agent understands her own community well and could determine whether the candidate is winning if she grasped his stances and policies—and she would grasp them accurately were they conveyed in the form she assimilates best. Agents differ in this respect: some absorb audio-visual material most readily, some long-form articles backed by facts, figures, and citations, some only brief and focused content, and some only their native language.

The campaign sends digital advertisements (private signals) to agents at varying precisions, where precision encodes the degree to which content is tailored to the agent’s mode of comprehension. A precise signal conveys the true state faithfully, helping the agent understand the candidate’s policies, record, and character and thereby correctly infer whether he will win. Crucially, a precise signal need not induce support for the campaign’s

candidate: if the candidate is in fact losing in the agent’s community, a highly precise signal drives her to oppose him. An imprecise signal, by contrast, is likelier to induce backing, since by obscuring the true state—through vagueness or confusion—it inflates the chance that the agent believes the candidate is winning. The biased planner is thus incentivized to *reduce* an agent’s signal precision whenever it believes its candidate is losing. This is the concrete counterpart of the obfuscation phases of Theorem 5. Tailoring an advertisement requires researching how each agent best assimilates content and hence is costly; even rendering a signal deliberately imprecise is costly, as it too demands agent-specific tailoring—to lower precision by burdening an agent with long-form, verbose articles, for instance, the planner must first know that she prefers focused, brief messages. The planner selects the precisions that maximize the expected number of backers net of the cost of generating those precisions.

B SOCIAL LEARNING DYNAMICS

We first summarize the ordered sequence of events at step i :

1. The planner determines q_i as a function of $\mathcal{H}_i$, and q_i is observed by all agents.
2. Agent i receives a private signal s_i satisfying $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i selects the action a_i that maximizes her own utility, conditioning on the observable history and her private signal.
4. All remaining agents and the planner observe a_i , and the public belief is updated to b_{i+1} accordingly.

By step (1), q_i is a deterministic function of $\mathcal{H}_i$; consequently, conditioning any random variable on both q_i and $\mathcal{H}_i$ renders the conditioning on q_i redundant—a principle we invoke repeatedly in what follows.

B.1 PRIVATE BELIEF UPDATE

We now derive agent i ’s action a_i . Let $\alpha_i := \frac{b_i}{1-b_i}$ denote the public likelihood ratio implied by the observed actions. Agent i forms her private belief $\tilde{b}_i = \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i\}$ by combining the current public belief with her private signal s_i , and we work with the associated private likelihood ratio $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$. We compute $\tilde{\alpha}_i$, and hence $\tilde{b}_i$, as follows.

1. If private signal is $s_i = G$, then

$$\begin{aligned} \tilde{\alpha}_i &= \frac{\tilde{b}_i}{1-\tilde{b}_i} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B | \mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G | \omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B | \mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1-q_i} \frac{b_i}{1-b_i} \end{aligned} \tag{8}$$

$$= \frac{q_i}{1-q_i} \alpha_i \tag{9}$$

In deriving Equation (8) we invoke the conditional-independence assumption—that, given ω , each agent’s signal is independent of the others’ and of the history—to write $\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G | \omega = G\} = q_i$, and analogously when conditioning on $\omega = B$; we additionally use that q_i is a function of $\mathcal{H}_i$.

2. If the private signal is $s_i = B$, the same reasoning yields

$$\tilde{\alpha}_i = \frac{1-q_i}{q_i} \alpha_i \tag{10}$$

Let $y(b, q)$ and $z(b, q)$ denote, respectively, the probability of realizing signal G or signal B given public belief b , signal precision q , and the history. Conditioning on ω and applying the law of total probability yields

$$\begin{aligned} y(b_i, q_i) &= \mathbb{P}\{s_i = G | \mathcal{H}_i\} \\ &= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G | \mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k | \mathcal{H}_i\} \\ &= q_i b_i + (1-q_i)(1-b_i) = 1 + 2q_i b_i - q_i - b_i \end{aligned} \tag{11}$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \quad (12)$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) gives

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1-q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \quad (13)$$

The expression above is well-defined: whenever $s_i = G$ the denominator $y(b_i, q_i) = \mathbb{P}(s_i = G | \mathcal{H}_i)$ is strictly positive, and whenever $s_i = B$ the denominator $z(b_i, q_i)$ is likewise strictly positive.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

Agent i 's action a_i is G when $\tilde{b}_i > 0.5$ (equivalently $\tilde{\alpha}_i > 1$), B when $\tilde{b}_i < 0.5$ (equivalently $\tilde{\alpha}_i < 1$), and s_i at the tie, per the tie-breaking rule. This action, together with the precision q_i , is observed by all peers and induces the updated public belief.

Suppose first that $b_i < 1 - q_i$, so that $\alpha_i < \frac{1-q_i}{q_i}$. Then if $s_i = G$, Equation (9) gives $\tilde{\alpha}_i < 1$, while if $s_i = B$, Equation (10) gives $\tilde{\alpha}_i < \frac{(1-q_i)^2}{q_i^2} \leq 1$ because $q_i \geq 0.5$. Hence $a_i = B$ regardless of the signal realization whenever $b_i < 1 - q_i$, and a symmetric argument shows $a_i = G$ whenever $b_i > q_i$. In both cases the strength of the public belief overpowers the precision of the private signal.

Now suppose $1 - q_i \leq b_i \leq q_i$. If $s_i = G$, then $\tilde{\alpha}_i \geq 1$ by Equation (8) and $b_i \geq 1 - q_i$; if $s_i = B$, then $\tilde{\alpha}_i \leq 1$ by Equation (10) and $b_i \leq q_i$. Combined with the tie-breaking rule, this yields $a_i = s_i$, establishing Equation (2), which we reproduce below.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Agent i 's action is therefore informative only in the first case of Equation (2), where it lets future agents perfectly recover her private signal; here the public belief updates exactly as the private belief in Equation (13). In the remaining two cases the action carries no information about the true state, so the public belief is unchanged. Together these yield Equation (3), reproduced below:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (14)$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

We now prove that, for $a \in \{G, B\}$, the quantity $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ depends on the history only through b_i and q_i ; that is,

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \quad (15)$$

For the second and third cases of Equation (2) this is immediate, since the action is a deterministic function of b_i and q_i alone. It remains to treat the first case.

$$\begin{aligned} & \mathbb{P}(a_i = G | \mathcal{H}_i, q_i) \\ &= \sum_{w \in \{G, B\}} \mathbb{P}(\omega = w | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = w) \\ &= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\ &= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\ &= b_i q_i + (1 - b_i)(1 - q_i) \end{aligned} \quad (16)$$

The first equality is the law of total probability. The second uses that q_i is a function of $\mathcal{H}_i$ to substitute b_i for $\mathbb{P}(\omega = w | \mathcal{H}_i, q_i)$. The third uses that $a_i = s_i$ on $1 - q_i \leq b_i \leq q_i$ together with the conditional independence of s_i from the history given ω . The result follows.

This establishes the following theorem.

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G|b_i, q_i, a_i\} \quad (17)$$

Equivalently, b_i is a sufficient statistic: it captures all of the information in $\mathcal{H}_i$ that is relevant to the update of b_{i+1} .

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G|b_i, q_i\}}{\mathbb{P}\{a_i|b_i, q_i\}}$$

We will prove the claim via induction on i .

For our base case, consider $i = 1$ so that $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. An application of Bayes' Theorem provides the following:

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G|b_1, q_1\} \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\} \mathbb{P}\{\omega = G|b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \end{aligned}$$

where the final step follows as the unconditioned belief of ω is equal to the prior belief b_1 .

Now, assume the claim holds for all $i \leq n$.

$$\begin{aligned} b_{n+1} &= \mathbb{P}\{\omega = G|\mathcal{H}_n, a_n, q_n\} = \frac{\mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\} \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G|\mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\} \mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G|b_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and the definition of b_n .

This completes the proof. $\square$

The Markovianity of the public belief is thus established by adapting the arguments of Banerjee (1992) and Bikhchandani et al. (1992); the adaptation is necessitated by the heterogeneous agent precisions that the introduction of a central planner induces, a feature absent from the classical fixed-precision setting.

C PROOFS

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We prove this for each case of a_i delineated in Equation (2). In the second case, we can reason as follows:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq G|b_i, q_i) = 1 - b_i$$

Similarly, in the third case of Equation (2), we obtain:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq B|b_i, q_i) = b_i$$

For the case when $1 - q_i \leq b_i \leq q_i$ and $a_i = s_i$ from Equation (2), we can reason as follows:

$$\begin{aligned} \mathbb{P}(a_i \neq \omega | b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a | b_i, q_i) \mathbb{P}(\omega \neq a | b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G | b_i, q_i) \mathbb{P}(\omega \neq G | b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B | b_i, q_i) \mathbb{P}(\omega \neq B | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqn 2} \quad (18)$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B | b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqns 11, 12, Thm 6} \quad (19)$$

$$= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1 - q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \quad \text{Eqn 13} \quad (20)$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases}$$

Combining the three cases yields the claim. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Applying Equation (4) to $r_A(b, q)$ in Equation (5) allows us to reason as follows

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \quad (21)$$

As established in Section 3.2, monotonicity of social welfare in precision (Appendix C.9) permits restricting the altruistic planner to $q \in [p, 1]$, with $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) makes precision p optimal whenever $p < \max(b, 1 - b)$: when the signal is too imprecise to alter the agent's action, the planner expends nothing on it. In the complementary case, the increasing concavity of $\beta(\cdot)$ makes $\dot{\beta}(\cdot)$ positive and decreasing, so any optimum of $r_A(b, q)$ lying in this range must be attained at the endpoint $q = 1$.

We now determine the regime in which this latter case is optimal.

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1 - b, 1 - p) &< -\beta(1) \\ \min(b, 1 - b, 1 - p) &> \frac{\beta(1)}{C} \end{aligned}$$

When $1 - p \leq \frac{\beta(1)}{C}$ the inequality never holds and precision 1 is never optimal; otherwise it holds precisely for $b \in [\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C}]$. This establishes Theorem 1. $\blacksquare$

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch We prove by induction that the expected k -th-stage reward—the planner's expected utility from the control and action of the k -th agent—is convex in the public belief. The base case is the instantaneous reward Equation (5), which is convex in the belief.

The first obstacle is the unusual structure of the public-belief update: although the state space is uncountable, from any belief the update is supported on at most two values. To exploit this, we introduce the *decision tree*—the complete binary tree of all trajectories generated once an initial belief and policy are fixed—in which each node has two children, one per signal realization the next agent may receive. The root records the expected instantaneous reward at time 1, and the induction descends the tree level by level, the k -th level comprising the 2^{k-1} nodes indexed by the sequence of $k - 1$ preceding signal realizations.

We then show that, whenever a node at level $k - 1$ has convex expected reward, both of its level- k children inherit that property. This is precisely where the dependence of actions on the public belief must be confronted:

under identical precision and signal realization, two agents starting from different beliefs may take opposing actions (Equation (2)), so the standard tools for bounding future terms of a Markov process do not apply. Here our specific belief update becomes an asset: we exploit the martingale property of Bayesian updates, $\mathbb{E}[b_{i+1}] = b_i$, which together with the convexity of the instantaneous reward closes the inductive step and completes the proof.

Preliminaries We first record several useful properties of the optimal value function $V_A^*(\cdot)$ in the following lemma.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1-b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$ and let Λ_n and Λ_b be the decision trees rooted at states n_0 and b_0 , respectively. Showing that Λ_n and Λ_b are symmetric suffices to show the symmetry of $V^*(\cdot)$. This will be accomplished by showing that the expected rewards, belief state values, and branching probabilities of the trees are reflections of one another.

First consider the instantaneous reward function r_A . Note that the decision function of the agents when choosing their actions is symmetric from Equation (2), thus, $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$. Applying this, we obtain

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - Cn_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - Cb_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - Cb_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Thus, the instantaneous reward function is symmetric about 0.5.

Now consider the belief state update when beginning with belief n_i assuming signal precision q_i . we will use $\bar{x}$ and $\underline{x}$ to represent updated beliefs from x after observing action G or B , respectively. From Equation (13):

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \underline{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

Thus, the belief updates are also symmetric. Combining this with the symmetry of the reward we obtain:

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \underline{b}_i) = r(\underline{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Thus, the expected rewards and belief state values of Λ_n and Λ_b are mirror images of one another.

Finally, consider the branching probabilities which we will denote as $y(b, q) = 1 + 2bq - b - q$ for observing action G and $z(b, q) = 1 - y(b, q)$ for observing action B .

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

Thus, the branching probabilities of Λ_n and Λ_b are also symmetric, completing the proof.

Extremal values

We will show that $V_A^*(0) = 0$ and rely on the symmetry about 0.5 proven above to show that $V_A^*(1) = 0$.

From Equation (3), when $b_0 = 0$, $b_0^+ = b_0^- = 0$ for any value of precision. Thus, regardless of signal realizations or control actions, the agents' actions and, subsequently, the public belief, will remain unchanged. That is, $b_i = 0 \forall i$ and, from Equation (2), $a_i = B \forall i$.

From Equation (5), we can then write:

$$\begin{aligned}
V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - C b_i \mathbb{P}\{a_i = B | \omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G | \omega = B, b_i = 0\}] \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right]
\end{aligned} \tag{22}$$

Thus, to attain $V_A^*(0)$, it suffices to choose $\pi(0)$ such that $\beta(\pi(0))$ is minimized for each i . Recall that $\beta(\cdot)$ is increasing on $[0.5, 1]$ and takes its minimum at $\beta(p) = 0$. Thus, by choosing $q_i = p \forall i$, Equation 22 yields $V_A^*(0) = 0$ with corresponding optimal policy $\pi_A^*(0) = p$.

The aforementioned symmetry about 0.5, shows that $V_A^*(1) = 0$, completing the proof. $\square$

We now formalize the notion of a *decision tree* for both Markovian and general policies. Any policy is represented by a complete binary tree, as in figure 3, in which each node encodes the state of the system—the belief value at which the controller sends its private signal to the agent at that level—and each branch encodes the action the agent takes upon receiving the signal. The controller’s policy, i.e. its choice of precision, fixes both the probability of each action emanating from a node (the *weight* of the branch) and the belief resulting from that action, so a decision tree is uniquely determined by its node beliefs and branch weights. Under a deterministic stationary Markovian policy the precision, and hence the action probability, is a deterministic function of the node’s belief value alone—independent of the node’s level and of the path from the root—so the entire tree is pinned down by the initial state. Under an arbitrary policy, by contrast, the controller’s decision at a node may depend on the entire path leading to it.

Proof of Theorem 2 We now prove the convexity of $V_A^*(\cdot)$ with respect to public belief.

Proof.

We first show that, at any fixed precision, the instantaneous reward r_A is convex in the public belief. Using Lemma 7, we rewrite r_A as

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \tag{23}$$

The pointwise minimum of linear functions is concave, and since r_A subtracts this minimum, it is convex in x .

Fix arbitrary x_0, m_0, n_0 with $tm_0 + (1 - t)n_0 = x_0$ for some $t \in [0, 1]$. Establishing convexity of $V_A^*(\cdot)$ amounts to showing $V_A^*(x_0) \leq tV_A^*(m_0) + (1 - t)V_A^*(n_0)$. Since $V_A^*(x) \geq V_A^\pi(x)$ for every policy π by optimality, it suffices to exhibit a policy $\tilde{\pi}$ satisfying

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1 - t)V_A^{\tilde{\pi}}(n_0) \tag{24}$$

Let Λ denote the decision tree of π^* initialized at x_0 .

We take $\tilde{\pi}$ to be the policy that, at each node, applies the precision Λ applies at the corresponding *counterpart* node—two nodes being counterparts when they are reached by the same sequence of actions from their respective roots (see figure 3). Thus $\tilde{\pi}$ replicates the precision Λ chooses after any given sequence of agent actions. We write Λ_m and Λ_n for the decision trees of $\tilde{\pi}$ initialized at m_0 and n_0 , respectively.

We now describe the policy $\tilde{\pi}$ at initial values m_0 and n_0 through the decision trees Λ_m and Λ_n , exhibiting the counterpart nodes and branch weights along the way. Fix a single action path λ through these trees by letting $\mathcal{G}$ be the set of epochs at which action G is taken and $\mathcal{B}$ its complement (see figure 3). Considering the decision tree Λ , we will refer to x_i as the state at epoch i when beginning with belief x_0 and following π^* along this path of actions. That is, x_i assumes one of 2^i possible belief values at epoch i starting from x_0 . In decision tree Λ_m (respectively, Λ_n), we use m_i (n_i) to denote the public belief after beginning with m_0 (n_0), following the path λ . Node m_i (respectively n_i) in decision tree Λ_m (Λ_n) is the counterpart of node x_i from decision tree Λ . Note that, $\tilde{\pi}(m_i)$ can now be illustrated notationally. Specifically, $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$. Similarly, $\tilde{\pi}$ can be defined at each node of Λ_m , considering the action taken by π^* at each counterpart node in Λ . Note that $y(x_i, \pi_A^*(x_i))$ (respectively, $z(x_i, \pi_A^*(x_i))$) is the weight of the branch corresponding to G (B) emanating from x_i in Λ . Because $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$ is the weight of the branch emanating from m_i for action G in Λ_m . This extends similarly to branches associated with action B and Λ_n .

Thus, the three decision trees differ in both the values of the nodes and the weights of the branches. Since π^* is deterministic, so is $\tilde{\pi}$. Therefore, from Equation (13) and Equation (3), x_i , m_i , and n_i are deterministic quantities once the specific path of actions is specified by $\mathcal{G}$.

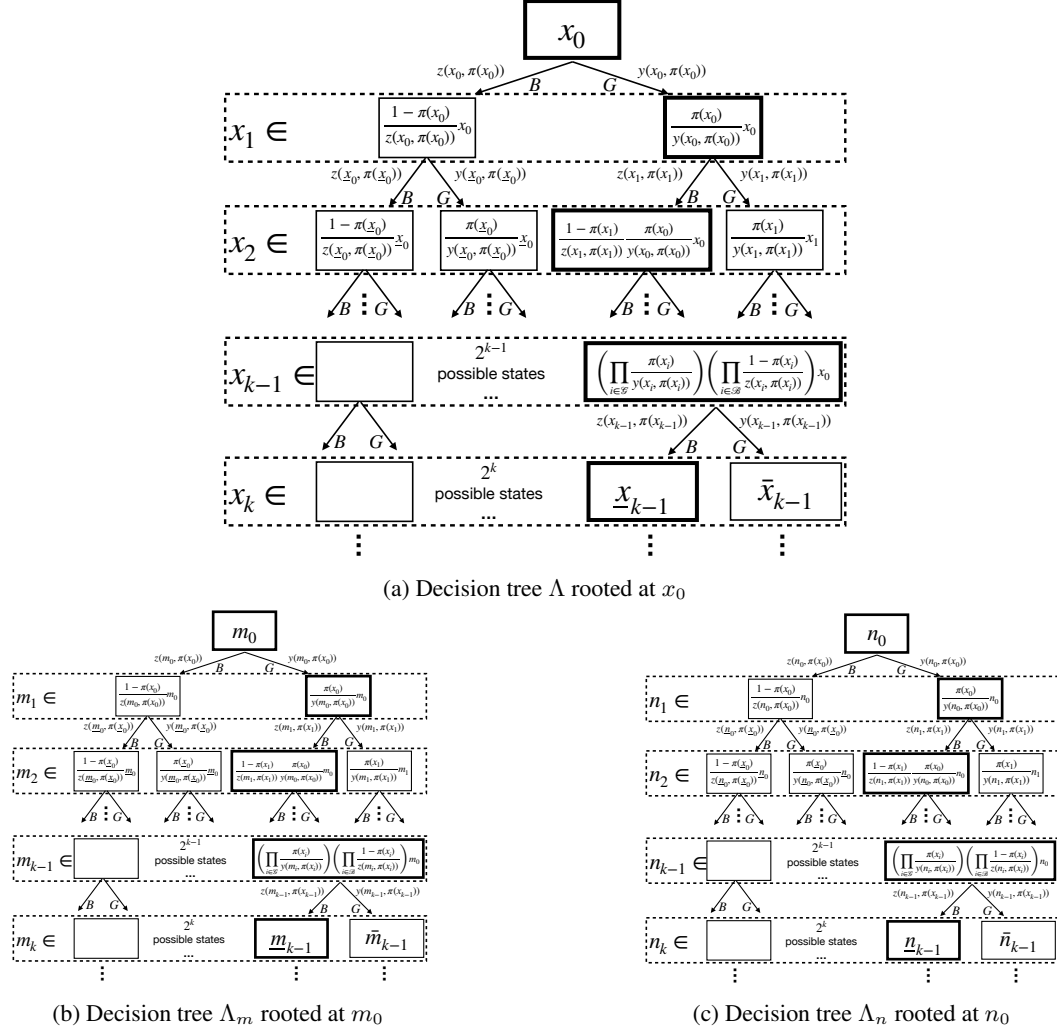


Figure 3: Belief evolution depicted as three binary trees rooted at states x_0 , m_0 , and n_0 , respectively, where level i contains the 2^i states reachable after i epochs. Left branches correspond to action B , yielding the updated belief x_0 (respectively $\underline{m}_0$, $\underline{n}_0$); right branches correspond to action G , yielding $\bar{x}_0$ (respectively $\bar{m}_0$, $\bar{n}_0$). Each branch is labeled with its probability, and the bold states trace an example path λ in which $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$. Bold states at each level of the three trees are counterparts of one another.

Now let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ be the expected instantaneous reward at epoch k when beginning with state b_0 and applying policy π and $\phi^*(x) = \phi^{\pi^* A}(x)$. Applying Fubini's Theorem, we can rewrite our value function as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

Thus, we can rewrite (24), as

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^*(m_0) \right] + (1-t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^*(n_0) \right] \quad (26)$$

To prove this, we show that for all $i = 0, 1, 2, \dots$

$$\phi_i^*(x_0) \leq t \phi_i^*(m_0) + (1-t) \phi_i^*(n_0) \quad (27)$$

leading directly to (26), (24), and finally the convexity of $V^*(\cdot)$ with respect to public belief.

Now let us consider general $i = k$. $\phi_{k-1}^\pi(x_0)$ will have 2^{k-1} terms each corresponding to one path of possible actions in the set $\{G, B\}^{k-2}$. Thus, at epoch $k-1$, the path λ will yield the following term in $\phi_{k-1}^*(x_0)$

$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

We now define $P_\lambda^*(x_0) = \left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right)$, that is, the probability of following the specified path λ when beginning with state x_0 and following policy π_A^* . Again, we will use $\bar{x}_i$ and $\underline{x}_i$ to represent updated beliefs from x_i after observing action G or B , respectively. Thus, $\bar{x}_i$ and $\underline{x}_i$ constitute belief values of the children nodes (in the $(i+1)$ th level) of the node with belief value x_i in the i -th level. Refer to levels $k-1$ and k in figure 3a. Considering the children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ (in the k -th level) of the node x_{k-1} (in the $(k-1)$ -th level), we obtain the following two terms in $\phi_k^*(x_0)$:

$$P_\lambda^*(x_0) [y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (29)$$

Similarly, we define $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)) \right)$, that is, the probability of following the specified path λ when beginning with state m_0 and following policy $\tilde{\pi}$. Because $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)) \right)$. We also define $P_\lambda^{\tilde{\pi}}(n_0)$ by replacing $\{m_i\}$ with $\{n_i\}$.

The corresponding terms in $\phi_k^{\tilde{\pi}}(m_0)$ and $\phi_k^{\tilde{\pi}}(n_0)$ are

$$P_\lambda^*(m_0) [y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (30)$$

$$P_\lambda^{\tilde{\pi}}(n_0) [y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (31)$$

We will show that

$$\begin{aligned} & P_\lambda^*(x_0) [y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\ & \leq t P_\lambda^*(m_0) [y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\ & \quad + (1-t) P_\lambda^{\tilde{\pi}}(n_0) [y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \end{aligned} \quad (32)$$

The same construction combines each sibling pair descending from every term at level $k-1$, producing an analogous inequality in each case; summing these inequalities yields (27), and hence (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We now focus on the latter of the two terms of (29). We will show the following two properties

$$\begin{aligned} \underline{x}_{k-1} &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ & \quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned} 1 &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\ & \quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \end{aligned} \quad (34)$$

Defining $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, it follows from (33) and (34) that

$$\underline{x}_{k-1} = \tilde{t} \underline{m}_{k-1} + (1-\tilde{t}) \underline{n}_{k-1} \quad (35)$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, it follows that:

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = t z(m_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(m_0) \quad (36)$$

Similarly, we can show

$$\tilde{t}z(x_{k-1}, \pi_A^*(x_{k-1}))P_\lambda^*(x_0) = (1-t)z(n_{k-1}, \pi_A^*(x_{k-1}))P_\lambda^{\tilde{\pi}}(n_0) \quad (37)$$

We apply this as follows to the latter term of (29):

$$\begin{aligned} & P_\lambda^*(x_0)z(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\ & \leq P_\lambda^*(x_0)z(x_{k-1}, \pi_A^*(x_{k-1})) \\ & \quad [\tilde{t}r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t})r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{Eqn 35}) \\ & = t \left[P_\lambda^{\tilde{\pi}}(m_0)z(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & \quad + (1-t) \left[P_\lambda^{\tilde{\pi}}(n_0)z(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (\text{Eqns 36, 37}) \end{aligned} \quad (38)$$

Applying the same argument to the first term of (29) yields:

$$\begin{aligned} & P_\lambda^*(x_0)y(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\ & \leq t \left[P_\lambda^{\tilde{\pi}}(m_0)y(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \right] \\ & \quad + (1-t) \left[P_\lambda^{\tilde{\pi}}(n_0)y(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \right] \end{aligned} \quad (39)$$

Summing (38) and (39) yields (32).

It remains only to prove (33) and (34), which together complete the proof of (32) and hence the theorem.

Proof of (33)

From the recurrences Equation (13) and Equation (3), we express $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as products of the successive belief updates applied to x_0 , m_0 , and n_0 , respectively, each factor being the multiplicative state update induced by a single action:

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \quad (40)$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \quad (41)$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \quad (42)$$

These relations are most readily understood through the illustration in figure 3. Reorganizing terms and using $x_0 = tm_0 + (1-t)n_0$, we relate $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$:

$$\begin{aligned} \underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ & \quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \end{aligned} \quad (43)$$

This proves (33).

Proof of (34)

Consider the numerator of the first term in (34), namely $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We prove that it can be written as $\alpha_0 m_0 + \beta_0$, where α_0, β_0 depend only on $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$ with functional form determined by the action sequence along the path in stages 1 through $k-1$. The proof proceeds by

backward induction on the terms of the product, starting from the $(k-1)$ -th. The base case is thus this final term, which is either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$ according to the action taken at stage $k-1$:

$$\begin{aligned} y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1})) \end{aligned}$$

Hence the final term has the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$, where α_{k-1} and β_{k-1} depend only on $\pi_A^*(x_{k-1})$ and their functional form is fixed by the $(k-1)$ -th action (which determines whether the term is $y(\cdot)$ or $z(\cdot)$).

For the inductive step, suppose $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i))$ equals $\alpha_{j+1}m_{j+1} + \beta_{j+1}$ for coefficients $\alpha_{j+1}, \beta_{j+1}$ that depend only on $\{\pi_A^*(x_i)\}_{i > j}$, with form determined by the actions along the path from $i > j$ to $i = k-1$. We show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i))$ equals $\alpha_j m_j + \beta_j$, with α_j, β_j depending only on $\{\pi_A^*(x_i)\}_{i > j-1}$ and form determined by the actions from $i > j-1$ to $i = k-1$. Defining $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and $g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}$, we obtain

$$\begin{aligned} &\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1}) \end{aligned}$$

Thus, from the definitions of $f(\cdot), g(\cdot), y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In either case the inductive hypothesis makes $\alpha_{j+1}, \beta_{j+1}$ depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the actions from $i > j$ to $i = k-1$, so the claim holds for α_j, β_j , completing the induction.

The denominator of (34), $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$, involves the same policy and action sequence as the numerator. Since the coefficients α_0, β_0 depend only on $\{\pi_A^*(x_i)\}$ and the action path, the identical argument writes the denominator as $\alpha_0 x_0 + \beta_0$ for these same α_0, β_0 , and likewise expresses the numerator of the second term of (34). The coefficients in (33) therefore satisfy

$$\begin{aligned} &t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\ &= \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\ &= \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = tm_0 + (1-t)n_0) \\ &= 1 \end{aligned}$$

Thus, (34) holds, completing the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Suppose, for contradiction, that $\pi_A^0(b) > \pi_A^*(b)$ for some $b \in [0, 1]$. We exhibit a policy π for which $V_A^*(b) < V_A^\pi(b)$, contradicting the optimality of $V_A^*(\cdot)$.

Let π apply precision $\pi_A^0(b)$ in the current step, starting from public belief b , and follow the optimal policy thereafter. Decomposing $V_A^*(b) - V_A^\pi(b)$ into the instantaneous reward r_A and the discounted future reward, and writing b^+ , b^- for the beliefs updated after signals G , B under the optimal policy and b_m^+ , b_m^- for the corresponding updates under the myopic policy, we obtain

$$\begin{aligned} V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\ &\quad + \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\ &\quad - \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\ &< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\ &\quad - \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\ &\leq 0 \quad (\text{Theorem 2}) \end{aligned}$$

The first inequality uses that, by definition of the myopic optimal policy, the instantaneous reward under $\pi_A^0(\cdot)$ is at least that under $\pi_A^*(\cdot)$. The final inequality invokes the convexity of $V_A^*(\cdot)$ from Theorem 2: since Equation (13) gives the ordering $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$, convexity bounds the discounted future term as required, which completes the proof. ■

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Let $V_A^\pi(b)$ denote the value of the policy π that applies precision q in the current step and follows the optimal policy thereafter; the maximizer q of $V_A^\pi(b)$ is then the optimal precision at public belief b . Because the planner never benefits from lowering precision below the baseline p (Appendix C.9), we restrict q to $[p, 1]$ without loss of generality.

Writing b^+ and b^- for the positive and negative belief updates following a signal of precision q , the second derivative of $V_A^\pi(b)$ with respect to q , for $q \geq \max(b, 1 - b)$, is

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2(1 - b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^3(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^3(b, q)} \right] & q \geq \max(b, 1 - b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

By the convexity of $V_A^*(\cdot)$ and the concavity of $\beta(\cdot)$, this second derivative is non-negative, so $V_A^\pi(b)$ is convex in q on $[\max(b, 1 - b), 1]$ and is therefore maximized at one of the two endpoints of that interval. On $q \in [0.5, \max(b, 1 - b))$ the derivative of $V_A^\pi(b)$ in q is negative, so cost minimization selects the baseline p . The only candidate optimal precisions are thus p , $\max(b, 1 - b)$, and 1—the three phases of Theorem 3.

The expected value under the latter two candidates is always strictly negative, whereas under precision p it equals 0 at $b = 0$ and $b = 1$. Combined with the non-positivity and symmetry of the value function (Lemma 8), this guarantees a threshold d_A with $\pi_A^*(b) = p$ for $b \in [0, d_A) \cup (1 - d_A, 1]$, and Theorem 1 together with Lemma 9 places $d_A \in (0, t_M)$, with t_M as in Theorem 1.

The same two results further guarantee a threshold $t_A \in [d_A, t_M]$ with $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$, completing the proof. ■

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Substituting the action rule (2) into the instantaneous reward (6) yields the case decomposition

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Observe that $z(b, q) = b + q - 2bq$ is increasing in q when $b < 0.5$ and decreasing in q when $b > 0.5$.

At $b = 0.5$ we have $z(0.5, q) = 0.5$ for every precision, and since $q \geq 0.5 = \max(b, 1 - b)$ the first case of (44) applies. There the second term of $r_B(b, q)$ is constant in q , so the planner maximizes only the first term, which is achieved at $q = p$.

We now establish a sequence of lemmas that support the proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable everywhere except at $q = p$, and away from that point its derivative in the chosen precision is

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) & q \geq \max(b, 1 - b) \\ -C(1 - 2b) & \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) & \text{otherwise} \end{cases} \quad (45)$$

In the former case the second derivative of the reward, for $q \neq p$, is

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

which is non-negative by the concavity of $\beta(\cdot)$, so $r_B(b, q)$ is convex in q . The reward is therefore maximized either at one of the endpoints $\{\max(b, 1 - b), 1\}$ or at the non-differentiable point $q = p$. $\square$

Lemma 11.

For $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Since $p > 0.5$, we have $\max(b, 1 - b) = 1 - b > p$ for $b \in [0, 1 - p)$.

First suppose $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$. By (44) the reward is then maximized by minimizing $|q - p|$, so the best precision below $\max(b, 1 - b)$ is p .

Now suppose $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$. Lemma 10 gives $\pi_B^0(b) \in \{1 - b, p, 1\}$, and it remains only to eliminate $q = 1$. Comparing $r_B(b, 1)$ and $r_B(b, 1 - b)$ —both in the first case of (44)—the increasing $\beta(\cdot)$ and $p < 1 - b \leq 1$ make the first reward term larger at $1 - b$ than at 1, while the monotonicity of $z(b, q)$ in q for $b < 0.5$ gives $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$. Precision 1 is thus ruled out, completing the proof. $\square$

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. The term $-\beta(|q - p|)$ renders the reward strictly negative for every $q \neq p$. For $b \in (p, 1]$ and $q = p$ we have $p < \max(b, 1 - b) = b$, placing us in the third case of (44) with reward 0. Hence the optimal precision is p . $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. When $p = 0.5$ the interval $[1 - p, 0.5]$ reduces to $\{0.5\}$, where $p = 1 - b$ and thus $\pi_B^0(b) = p = 1 - b = 0.5$. For the remainder we take $p > 0.5 \geq b$.

For $b \in [1 - p, 0.5]$ we have $\max(b, 1 - b) = 1 - b \leq p$, so the first case of (44) gives $r_B(b, p) = -Cz(b, p)$. We distinguish $q < \max(b, 1 - b)$ from $q \geq \max(b, 1 - b)$.

If $q < \max(b, 1 - b) = 1 - b$, the second case of (44) applies with reward $-\beta(|q - p|) - C$. Because $z(b, q)$ is increasing in q for $b < 0.5$ and ranges over $[z(b, 0.5), z(b, 1)] = [0.5, 1 - b]$, it satisfies $z(b, q) < 1$ throughout; hence the reward from any $q < 1 - b$ is strictly below $r_B(b, p)$, ruling out this case.

If $q \geq \max(b, 1 - b)$, Lemma 10 leaves the candidates $\{1 - b, p, 1\}$. Comparing precision 1 to precision p , the first reward term of (44) is larger at p , and the monotonicity of $z(b, q)$ in q for $b < 0.5$ makes the second term larger at p as well. Precision 1 is therefore eliminated, completing the proof. $\square$

We now define the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

By definition, any optimal policy is also ϵ -optimal for every $\epsilon > 0$; in particular, $\pi_B^0(b)$ is ϵ -optimal.

We let $\pi_B^{\epsilon, 0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

- 1512 1. $\pi_B^0(b) \in \{p, 1\}$, or
 1513
 1514 2. $\pi_B^{\epsilon, 0}(b) = b - \epsilon$
 1515

1516 *Proof.* For $b \in (0.5, p]$ we have $\max(b, 1 - b) = b \leq p$, and we distinguish $q \geq \max(b, 1 - b)$ from
 1517 $q < \max(b, 1 - b)$.

1518 In the first case Lemma 10 gives $\pi_B^0(b) \in \{b, p, 1\}$. Comparing precision b to precision p , the first reward term
 1519 of (44) is larger at p , and since $z(b, q)$ is decreasing in q for $b > 0.5$ the second term is larger at p too; precision
 1520 b is thus eliminated, yielding the first claim.

1521 In the second case $q < \max(b, 1 - b) = b$, so $q \in [0.5, b)$ and the third case of (44) applies with reward
 1522 $-\beta(|q - p|)$. Because $0.5 < b \leq p$, this reward is increasing in q over $[0.5, b)$, which yields the second claim
 1523 directly. $\square$

1524 **Remark 3.** As noted in Puterman (1990), Section 2.3.1, over infinite action spaces an optimal policy may fail
 1525 to exist because the supremum is not attained. In that event we instead seek ϵ -optimal policies—those yielding
 1526 reward within $\epsilon > 0$ of the supremum—as arises specifically in case 2 of Lemma 14.

1527 **Proof of Theorem 4** Using $p \geq 0.5$, we treat the myopic biased policy in four cases: $b \in [0, 1 - p)$,
 1528 $b \in [1 - p, 0.5]$, $b \in (0.5, p]$, and $b \in (p, 1]$. By Lemmas 11–14, the candidate policies in these cases are,
 1529 respectively, $\{p, 1 - b\}$, $\{1 - b, p\}$, $\{1, p, b - \epsilon\}$, and $\{p\}$.
 1530

1531 Case 1: $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$ (Lemma 11)

1532 We establish a threshold t_1 with $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

1533 Applying (44), the rewards under the two candidate policies are

$$\begin{aligned} 1534 \quad r_B(b, p) &= -C \\ 1535 \quad r_B(b, 1 - b) &= -\beta(1 - b - p) - C(1 - 2b(1 - b)) \end{aligned}$$

1536 The derivative of $r_B(b, 1 - b)$ with respect to b is

$$1537 \quad \frac{\partial}{\partial b} r_B(b, 1 - b) = \dot{\beta}(1 - p - b) + 2C(1 - 2b)$$

1538 which is positive since $\beta(\cdot)$ is increasing and $b < 1 - p \leq 0.5$. Evaluating at the endpoints $b = 0$ and $b = 1 - p$
 1539 gives

$$\begin{aligned} 1540 \quad r_B(0, 1) &= -\beta(1 - p) - C < r_B(b, p) \\ 1541 \quad r_B(1 - p, p) &= -C(1 - 2p(1 - p)) > r_B(b, p) \end{aligned}$$

1542 Thus $r_B(b, 1 - b)$ increases on $[0, 1 - p)$, lying below $r_B(b, p)$ at $b = 0$ and above it at $b = 1 - p$. Both reward
 1543 functions are continuous (as $\beta(\cdot)$ is), so a threshold $t_1 \in (0, 1 - p)$ exists with $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and
 1544 $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.
 1545

1546 Case 2: $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{1 - b, p\}$ (Lemma 13)

1547 If $p = 0.5$ the interval reduces to $\{0.5\}$, where $1 - b = p$; the two candidates coincide and $\pi_B^0(b) = p$. For the
 1548 remainder we take $p > 0.5$.

1549 We prove the existence of thresholds t_2, t_3 with $\pi_B^0(b) = 1 - b$ for $b \in [t_2, t_3)$ and $\pi_B^0(b) = p$ for $b \in$
 1550 $[1 - p, t_2) \cup [t_3, 0.5]$.

1551 Applying (44), the rewards under the candidate policies are

$$\begin{aligned} 1552 \quad r_B(b, 1 - b) &= -\beta(p - 1 + b) - Cz(b, 1 - b) \\ 1553 \quad r_B(b, p) &= -Cz(b, p) \end{aligned}$$

1554 The optimal policy equals $1 - b$ precisely when

$$\begin{aligned} 1555 \quad r_B(b, 1 - b) &\geq r_B(b, p) \\ 1556 \quad \beta(p - 1 + b) &\leq C[z(b, p) - z(b, 1 - b)] \\ 1557 \quad \beta(p - 1 + b) &\leq C(p - 1 + b)(1 - 2b) \end{aligned} \tag{46}$$

1558 The left-hand side of (46) is a concave, increasing function of b on $[1 - p, 0.5]$, while the right-hand side is
 1559 a concave parabola that increases on the first half of the interval and decreases on the second. Partition the
 1600 interval into $[1 - p, 0.75 - 0.5p]$ and $[0.75 - 0.5p, 0.5]$. On the first half both sides are increasing and concave

and the right-hand side is a parabola, so they cross at most twice (once at $b = 1 - p$); on the second half the left-hand side increases while the right-hand side decreases, so they cross at most once. Denote the intersection points for $b > 1 - p$ by $t_2 \leq t_3$. It remains to identify the optimum on each of $(1 - p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, $p > 0.5$ gives $r_B(b, p) > r_B(b, 1 - b)$, so $\pi_B^0(b) = p$ on $(t_3, 0.5)$; consequently $\pi_B^0(b) = 1 - b$ on (t_2, t_3) and $\pi_B^0(b) = p$ on $(1 - p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b - \epsilon\}$ (Lemma 14)

Here we establish thresholds $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b - \epsilon$, respectively.

Applying (44), the rewards under the candidate policies are

$$\begin{aligned} r_B(b, 1) &= -\beta(1 - p) - C(1 - b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b - \epsilon) &= -\beta(p - b + \epsilon) \end{aligned}$$

At $b = p$, $r_B(b, b - \epsilon)$ is arbitrarily close to 0, so $b - \epsilon$ is optimal on some interval $(t_5, p]$ with $0.5 \leq t_5 < p$.

Since $r_B(b, 1)$ and $r_B(b, p)$ are linear in b while $r_B(b, b - \epsilon)$ is convex (by the concavity of β), the differences $r_B(b, b - \epsilon) - r_B(b, p)$ and $r_B(b, b - \epsilon) - r_B(b, 1)$ are convex and hence have convex level sets. The set on which both are positive—exactly the set on which $b - \epsilon$ is optimal—is therefore convex, so $b - \epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

It remains to compare precisions 1 and p on $(0.5, t_5]$, for which we examine the values of $r_B(b, 1)$ and $r_B(b, p)$ at 0.5 together with their derivatives:

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Since $r_B(0.5, 1) < r_B(0.5, p)$, there is an interval $(0.5, t_4)$ with $t_4 \leq t_5$ on which $\pi_B^0(b) = p$. As both functions are linear in b , their ordering changes at most once, so whenever $t_4 < t_5$ we have $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This is immediate from Lemma 12, completing the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. *For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,*

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. Applying (44), the derivative of the biased reward in b is

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

which, since $q \geq 0.5$ and $z(b, q) \in [0, 1]$, shows that $r_B(b, q)$ is non-decreasing in b .

Next, the Bayesian nature of the belief updates makes the belief process a martingale: $\mathbb{E}[b_{i+1}] = b_i$ for every fixed q_i . We verify this in two cases. If $q_i < \max(b_i, 1 - b_i)$, then (3) gives $b_{i+1} = b_i$, and the claim holds trivially. If $q_i \geq \max(b_i, 1 - b_i)$, then (13) supplies the distribution of b_{i+1} , from which the same identity follows.

The monotonicity of $V_B^*(\cdot)$ is then a consequence of Proposition 5 of Smith & McCardle (2002). □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first establish continuity of the optimal value function.

Lemma 16. *Biased value function continuity*

$$V_B^*(\cdot) \text{ is continuous w.r.t. } b.$$

Proof. Consider the one-stage value function $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. By Theorem 4, $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$, so we may write it equivalently as

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1-b, 1, b-\epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

Using the above expression, we now argue that $V_B^0(\cdot)$ is continuous on $b \in [0, 1]$. First, consider $b \leq 0.5$. From Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$. In either case, the instantaneous reward will not depend upon ϵ , and the supremum in (48) can be omitted, leaving $V_B^0(b) = \max_{q \in \{1-b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ falls in case one of (44) and is thus continuous with respect to b . On the same interval, $r_B(b, p)$ has one discontinuity at $b = 1 - p$ where it switches from case two to case one of (44). At $b = 1 - p$, Theorem 4 shows that the optimal myopic policy switches from $1 - b$ to p . Both policies fall in the first case of (44), and, at $b = 1 - p$, $p = 1 - b$, so the controls are equal. Thus, the left and right limits coincide with the optimal value, and $V_B^0(\cdot)$ is continuous.

For $b \geq 0.5$, Theorem 4 gives the candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained for $q = b - \epsilon$, the resulting value is $V_B^0(b) = \sup_{\epsilon > 0} r_B(b, b - \epsilon)$. From the third case of (44), we can rewrite this as $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$. Since $b - \epsilon$ is only selected for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Thus, exchanging the order of the maximum and supremum in (48), we obtain $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. For the first expression in this maximum, $r_B(b, 1)$ falls into case one of (44) and is thus continuous. The continuity of $-\beta(|p - b|)$ follows from the continuity of $\beta(\cdot)$. For $r_B(b, p)$, since $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third case applies for $b > p$. Thus, $r_B(b, p)$ is potentially discontinuous at $b = p$. Therefore, what remains is to show the continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, the myopic ϵ -optimal policy switches from $q = b - \epsilon$ to $q = p$ by Theorem 4 with $\pi_B^{\epsilon, 0}(p) = b - \epsilon$. Thus, for $b = p$, the inner maximum of (48) is attained at $q = b - \epsilon$ and, from the third case of (44), $V_B^0(p) = \sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Similarly, from Theorem 4, for each $\epsilon > 0$, there exists $t_5 < p$ such that on $[t_5, p]$ the inner maximum of (48) is attained by $q = b - \epsilon$. On this interval, $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$. Therefore, $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimal policy is $q = p$ from Theorem 4, and this yields reward $V_B^0(b) = 0$ for all $b > p$ from the third case of (44). This proves the continuity of $V_B^0(\cdot)$ at $b = p$ and completes the proof that $V_B^0(\cdot)$ is continuous on $[0, 1]$.

Applying Theorem 4.2 of Hinderer (2005), then yields the continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ in the public belief and the compactness of the control set $[0.5, 1]$, one might expect an optimum to exist at every public belief. An important subtlety forecloses this conclusion: while $V_B^*(\cdot)$ is continuous in the public belief, it need not be continuous in the precision. Concretely, b_{i+1} is a function of both b_i and q_i (see Equation (3)), so $\mathbb{E}[V_B^*(b_{i+1})]$ may be written as $\phi(b_i, q_i)$; because b_{i+1} need not be continuous in q_i , neither need $\phi(\cdot)$ be. There may therefore exist regimes in which the optimal value is attained by no policy—and we show below that such regimes do in fact occur.

Proof of Theorem 5 We now leverage Lemma 16 to prove Theorem 5. As before, let $z(b, q) = b + q - 2bq$ denote the probability that an agent with prior public belief b receives signal B under precision q , and $y(b, q) = 1 - z(b, q)$ the corresponding probability of signal G . Abusing notation, we write $V_B(b, q)$ for the expected utility of beginning at public belief b , applying precision q in the current step, and following the optimal policy $\pi_B^*(\cdot)$ thereafter. Applying (13), (3), and (44), we obtain

$$V_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) \\ \quad + \delta y(b, q) V_B^*(f(b, q, G)) \\ \quad + \delta z(b, q) V_B^*(f(b, q, B)) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) \\ \quad - C\mathbf{1}(b < 0.5) \\ \quad + \delta V_B^*(b) & \text{otherwise} \end{cases} \quad (49)$$

Consider the latter case of (49), where $q < \max(b, 1 - b)$. Here the only term affected by the precision policy is $-\beta(|q - p|)$, maximized by choosing q as close to p as possible. Hence, whenever the optimal precision falls below $\max(b, 1 - b)$, it lies as close to p as the constraint permits.

We now consider four cases: (1) $b \in [0, 1 - p)$, (2) $b \in [1 - p, 0.5]$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p)$, claims (A) and (B)

In this case, we will prove the existence of t_1 such that $\pi_B^*(b) = p$ for $b \leq t_1$. First note that if $\pi^*(b) < \max(b, 1 - b)$, we fall into the second case of Equation (49). Here, the only term affected by the precision

policy is $-\beta(|q - p|)$, which is maximized by minimizing $|q - p|$. Because $p \in [0.5, 1]$, when $b \in [0, 1 - p)$, $p < \max(b, 1 - b)$. Thus, for $b \in [0, 1 - p)$, if $\pi_B^*(b) < \max(b, 1 - b)$, then $\pi_B^*(b) = p$.

Now, we evaluate the value function at belief $b = 0$. At $b = 0$, the state of the world is known to be B and, thus, no signal, regardless of its precision, will move the public belief. This can be seen mathematically by substituting $b = 0$ into the updated belief expressions in (13). Furthermore, at $b = 0$, if $q \geq \max(b, 1 - b)$, then $q = 1$. As argued above, if the precision is less than $\max(b, 1 - b)$, the optimal precision is $q = p$. We can compare the resultant value under these two candidate policies as follows from Equation (49):

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Because $V_B(0, 1) < V_B(0, p)$, $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$. Thus, from Equation (51), $V_B^*(0) = \frac{-C}{1-\delta}$. By Lemma 15, this is a lower bound for the value function at any public belief. Therefore, and since $V_B^*(b) \leq 0$, $|V_B^*(b)| \leq \frac{C}{1-\delta}$ for all $b \in [0, 1]$.

We now show that there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). Thus, in this interval the optimum control lies in the second case of Equation (49) so $\pi_B^*(b) < \max(b, 1 - b)$. It follows from our initial argument earlier in this case that $\pi_B^*(b) = p$ in this interval, which would complete the proof.

We define function $g(\cdot)$ to be the maximum of the first case of the RHS of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q - p|) - Cz(b, q) + \delta y(b, q) V_B^*(f(b, q, G)) + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, $f(b, q, G)$ and $f(b, q, B)$ are both continuous with respect to q from Equation (13). Thus, the maximum in Equation (52) must exist because it is of a function which is continuous with respect to q over a compact set $q \in [\max(b, 1 - b), 1]$.

We now take the limit of $g(\epsilon)$ as ϵ tends to 0. Because $q \in [\max(\epsilon, 1 - \epsilon), 1]$, when $\epsilon \rightarrow 0$, $q \rightarrow 1$ (inside the maximum in Equation (52)). We now reason about each of the four terms in Equation (52). As $\epsilon \rightarrow 0$, $q \rightarrow 1$ so that 1) from the continuity of $\beta(\cdot)$, $\beta(|q - p|) \rightarrow \beta(1 - p)$, 2) from the continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$, $y(\epsilon, q) \rightarrow 0$. Thus, the second term goes to C . Because $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term goes to 0. Finally, because $z(\epsilon, q) \rightarrow 1$ and $f(\epsilon, q, B) \rightarrow 0$ and since $V_B^*(\cdot)$ is continuous from Lemma 16, the last term approaches $\delta V_B^*(0)$. Thus,

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (53)$$

Applying precision p , however, yields the following from the second case of Equation (49):

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p - p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The last equality follows from the continuity of $V_B^*(\cdot)$ from Lemma 16.

Note that the expression in Equation (53) is strictly less than that in Equation (55). Thus, there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). The result follows.

Case 2: $b \in [1 - p, 0.5)$, Claim (C)

We now argue that for $b_i \in [1 - p, 0.5)$, $\pi^*(b_i) \geq 1 - b_i$, yielding claim (C). Consider a stationary policy $\pi(\cdot)$ such that $\pi(b_i) < 1 - b_i$. Note that, from Equation (3), the public belief will remain unchanged under such a policy. Thus, the same control will be applied at every future time step, and, from Equation (2), every agent will select action B . Using the facts that $1 - b_i \leq p$ in this belief range, $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the expected total discounted utility under such a policy as follows:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Thus, if this policy is applied, it will result in expected value $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1 - \delta}$. That is, the cost of every agent choosing action B and continuing to apply precision $\pi(b_i)$. This is strictly lower than the value function lower bound of $\frac{-C}{1 - \delta}$ established using Lemma 15 in Case 1 because $\pi(b_i) < 1 - b_i \leq p$. Therefore, if an optimal policy exists at this point, it must be the case that $\pi_B^*(b_i) \geq 1 - b_i$.

The remaining candidate policies are $q_i \in [1 - b_i, 1]$. Applying Bellman's principle we can write the following:

$$V_B^*(b_i) = \sup_{q_i \in [1 - b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Because $q_i \geq \max(b_i, 1 - b_i)$ and $b \in [1 - p, 0.5] \subset (0, 1)$, Equation (13) and Equation (3) imply that $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1 - q_i)}{z(b_i, q_i)}$. Both of these expressions are continuous with respect to q_i and $V_B^*(\cdot)$ is continuous by Lemma 16, thus the RHS of Equation (56) inside the supremum is continuous with respect to q_i . Therefore, in Equation (56), the supremum over a compact set $q_i \in [1 - b_i, 1]$ of a function which is continuous with respect to q_i must be attained, implying the existence of an optimal policy. Thus, $\pi_B^*(b_i) \geq 1 - b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, $q \geq b$ as we assumed $q \in [0.5, 1]$. This places us in the first case of Equation (49). Furthermore, $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous with respect to q for $b = 0.5$. Thus, the supremum over $q \in [0.5, 1]$ of $V(b, q)$ must be attained as it is of a continuous function of q over a compact set of q . Therefore, an optimal policy must exist at $b = 0.5$.

Now consider $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ such that $\pi'(b) < b$. From Equation (3) and Equation (2), applying such a policy will result in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. Using the facts that $b_i \leq p$, $\mathbb{P}(a_i = G | b_i, \pi'(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the total expected discounted utility under such a policy as follows:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus, the policy results in expected value $V_B^{\pi'}(b_i) = \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$. Note that, in this case, $V_B^{\pi'}(b_i)$ is an increasing function of $\pi'(b_i)$. Thus, the supremum of $V^*(\pi')_B(b_i)$ over $\pi'(b_i) \in (0.5, b)$ is not attained.

We now consider two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$, and (2) the complement. In the former case, we can argue as we did for $b = 0.5$ that the supremum must be attained and an optimal policy exists such that $\pi_B^*(b_i) \geq b_i$. In the latter case, as argued above, no optimal policy exists. Instead, for sufficiently small $\epsilon > 0$, there exists an ϵ -optimal policy such that $\pi_B^\epsilon(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

What remains is to show that there exists $t_2 \in (0.5, p)$ such that for $b_i \in (t_2, p]$ we fall into the latter case, yielding claim (F). We define functions $m(\cdot)$ and $n(\cdot)$ as follows corresponding to the resultant value from choosing precisions less than b or greater than or equal to b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b)} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking the limit of $m(\cdot)$ as $b \rightarrow p$ yields:

$$\lim_{b \rightarrow p} m(b) = 0 \quad (57)$$

When $q \geq b$ we fall into the first case of Equation (49). Thus we can expand $n(b)$ as follows:

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q)V_B^*(b, q) \\ &\quad + \delta z(b, q)V_B^*(b, q)] \end{aligned} \quad (58)$$

We will argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Since p is a fixed parameter strictly less than 1, this means from Equation (57) that $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, which implies the existence of an open interval $(t_2, p]$ where an optimal policy does not exist, yielding claim (F). We now argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Note that every term of Equation (58) is bounded above by 0, so it suffices to show that any one term, say the second term $-Cz(b, q)$, is upper bounded by $-C(1 - p)$. First note that $z(b, q) = q(1 - 2b) + b$ so $z(b, q)$ is decreasing in q since $b > 0.5$. Thus, $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$. The last inequality follows because $b \in (0.5, p]$. The claim follows since $p < 1$.

Along with Equation (57), this implies the existence of an open interval (t_2, p) where an optimal policy does not exist, yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider applying stationary policy $\pi(\cdot)$ such that $\pi(b) = p$. Because $b > p$, Equation (3) and Equation (2) imply that such a policy results in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. We can apply Equation (6) to write

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

Thus, $V_B^\pi(b) = 0$. Since this is also an upper bound of the optimal value function for any b , $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. *Precision Monotonicity of Social Welfare*

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. It is clear that the one-step welfare $-C \min(b_i, 1 - b_i, \pi(b_i))$ is increasing in $\pi(b_i)$.

Furthermore, note that the convexity of social welfare with respect to public belief follows from the same argument as in the proof of Theorem 2. This is because social welfare is equivalent to the altruistic value function in the special case where $\beta(\cdot) = 0$.

We can then follow the same argument as in the proof of Lemma 9 (Appendix C.4) to show that increasing precision increases welfare. □

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

Here we show that, for the altruistic planner, the problem we consider is equivalent to one involving two types of individuals with opposing preferences—a first, concrete step beyond the homogeneous-agent assumption.

Assume there are two types of agents, G and B , with the type of agent i denoted t_i . The first type receives unit reward when $a_i = \omega$ and no reward otherwise; the second receives unit reward when $a_i \neq \omega$ and no reward otherwise. Assume further that every agent knows its own type and the types of all others. All remaining aspects of the model are unchanged.

Agent i 's private belief update is given below. It is unchanged from the homogeneous case, since it depends only on the signal structure $\mathbb{P}(s_i = \omega) = q_i$ and not on the agent's preferences.

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_i q_i - b_i - q_i} b_i & s_i = G \\ \frac{1-q}{b_i + q_i - 2b_i q_i} b_i & s_i = B \end{cases} \quad (59)$$

Using Equation (59), agent i 's action can be written as follows with superscript indicating the type of agent i and ! indicating negation:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Because each agent's type is known to all, when $q_i \geq \max(b_i, 1 - b_i)$ every agent can perfectly infer agent i 's private signal; when $q_i < \max(b_i, 1 - b_i)$, agent i 's action is uninformative. The type therefore never obstructs inference of the signal.

Consequently, after observing agent i 's type t_i , precision q_i , and action a_i , the public belief is updated as follows:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

In this setting an altruistic planner seeks to induce type- G agents to take action ω and type- B agents to take action $!\omega$, since these are exactly the choices that maximize each type's own utility.

The planner’s instantaneous reward is then, with the superscript denoting the type of agent i ,

$$\begin{aligned}
r^G(b_i, q_i) &= -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega) \\
&= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases} \\
r^B(b_i, q_i) &= -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega) \\
&= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}
\end{aligned}$$

These two reward functions are identical and coincide with the reward in Equation (5), so the altruistic planner’s problem with heterogeneous but observable types reduces exactly to the homogeneous problem analyzed in the main text. ■

E EXPERIMENTS

E.1 EXPERIMENTAL SETUP

To simulate the controlled social-learning setting, we consider a sequence of agents deciding whether to buy a new model of car, each informed by observations of past actions and by an advertisement whose precision is selected by the planner. LLMs occupied three distinct roles: (1) the agents, (2) the planner, and (3) an oracle. Each agent was assigned a ‘profile’ encoding demographic information together with the agent’s car use cases and preferences. At the outset of each run we designated either the good or the bad car as the offering. The information sets and objectives of the planner and agents were exactly those of the model in Section 3.

The oracle discharged two functions: (1) it generated a generic message serving as the baseline-precision signal absent any intervention by the planner, and (2) once the planner had chosen a precision, it generated a message of the prescribed precision, customized to the profile of the target agent.

The oracle was provided with a fact sheet (safety rating, customer satisfaction, and the like) for each type of car together with the agents’ profiles; jointly, the fact sheet and profile supplied the material from which a tailored message of the planner’s prescribed precision could be generated. Further detail on the experimental setup and prompting appears in Appendix E.2.

Each agent made a binary decision on the basis of the observed history of previous decisions and its own private signal, with the objective of matching the true state.

The planner was given the same decision history and tasked with choosing a precision level for each agent’s message, along with the precision of a generic, untailored message that it could send at no cost. In the altruistic case the planner’s objective was to induce correct decisions; in the biased case, to induce agents to buy the car.

We compare the LLM simulations against exact analytic solutions of the planner’s MDP obtained via Juli-aPOMDP (Egorov et al., 2017).

E.2 PARAMETERS AND PROMPTS

E.2.1 INSTANCE PARAMETERS

Experiments were run using the LG AI EXAONE Deep 32B model (LG AI Research, 2025); extending the study to additional model families is left to future work, and the single-family scope is a stated limitation of the empirical results. We solve the model analytically and run the LLM simulations over the parameter grid in Table 1.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$
p	Baseline signal precision	$\{0.6, 0.7, 0.8, 0.9\}$
δ	Discount Factor	$\{0, 0.25, 0.5, 0.75, 1\}$

Table 1: Instance parameters swept in both the analytical and the LLM experiments. The cost of an incorrect action is fixed at $C = 1$; the control-cost coefficient k , baseline precision p , and discount factor δ each range over the listed values.

The car facts were constructed so that the good car was unambiguously good for every agent and the bad car unambiguously bad for every agent, removing preference heterogeneity as a confound. We reproduce the fact sheets for the GOOD and BAD cars, respectively, below.

```
fact_sheet["GOOD"] = [
    "EPA-rated range: 290 miles.",
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",
    "8-year / 100,000-mile battery warranty.",
    "NHTSA 5-Star overall safety rating.",
    "Cold-soak -5 °F keeps 93 %% range.",
    "Cools cabin 100→72 °F in 3 min.",
    "Cargo volume 26 ft3 total.",
    "Tow's 1 500 lb with factory hitch.",
    "Average 1.2 service visits first 3 yrs.",
    "92 %% owners would buy again (Apr-25).",
    "Qualifies for federal credit.",
    "Quarterly OTA updates incl. battery pre-heat."
]
fact_sheet["BAD"] = [
    "EPA-rated range: 155 mi (combined).",
    "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",
    "Battery warranty: 3 yr / 36 000 mi, first owner only.",
    "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
    "Cold-soak 15 °F retains 58 %% range; cabin heat cuts range 20%.",
    "Cabin cools 105 °F → 72 °F in 11 min.",
    "Cargo volume 17 ft3; rear seats fixed (no fold-flat).",
    "No factory towing approval.",
    "Average 3.8 service visits per year (battery-alert issues).",
    'Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
    "Not eligible for federal tax credit.",
    "OTA updates limited to infotainment."
]
```

Profiles were generated programmatically, each specifying an agent's name, age, city of residence, interests, planned use case for the car (commuting, road trips, and so on), and principal pain points or concerns (fuel efficiency, winter driving, and the like).

E.2.2 PROMPTS

Agent LLM Prompt:

```
sys = (
    f"You are deciding whether to buy a car. "
    f"Your profile is {profile}"
    f"You start believing there's a 50% chance the car is GOOD. "
    f"You have two information sources: "
    f"1. Previous agents' decisions "
    f"2. An advisor's message about the car "
    f"Your goal is to maximize your reward "
    f"i.e. BUY if the car is GOOD and "
    f"NOT BUY if the car is BAD. "
    f"If you BUY a GOOD car, you GAIN ${C}. "
    f"If you NOT BUY a BAD car, you GAIN ${C}. "
    f"If you BUY a BAD car, you LOSE ${C}. "
    f"If you NOT BUY a GOOD car, you LOSE ${C}. "
    f"belief = your probability (0.0-1.0) that the car is GOOD. "
    f"decision = exactly 'BUY' or 'NOT BUY'."
)
user = (
    f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
    f"ADVISOR MESSAGE:\n{message}\n\n"
    f"TASK: Use the previous decisions and the advisor message. "
    f"Decide BUY or NOT BUY. "
    f"Update your belief about car quality (0.0-1.0). \n\n"
    f'JSON response:{{"belief": <number>,"decision": "<BUY or NOT BUY>"}}'
```

```

1944 )
1945
1946 Planner LLM Prompt:
1947
1948 sys = (
1949     f"You are providing messages to potential car buyers. "
1950     f"Each potential car buyer (or agent) has the same information. "
1951     f"Your goal is to get agents to {objective}."
1952     f"Each decision according to your objective, earns you {C} dollars. "
1953     f"You and the agents have the same information. "
1954     f"Initially there is a 50% chance the car is GOOD. "
1955     f"The agents' aim to BUY if the car is GOOD "
1956     f"and NOT BUY if the car is BAD. "
1957     f"You can send a baseline message for free "
1958     f"or tailor the message to the agent at a cost. "
1959     f"Tailoring costs {k} dollars times the change in precision. "
1960     f"(Message precision is the probability (0.5 to 1.0) "
1961     f"that the message leaves the buyer with the correct impression. "
1962     f"For example, if the car is likely GOOD: "
1963     f"a high precision message will likely suggest BUY "
1964     f"and a low precision message will likely suggest NOT BUY. "
1965     f"You may choose to increase or decrease precision. "
1966     f"Agents decide based on your message and the decision history. "
1967     f"They update beliefs by observing previous decisions. "
1968     f"IMPORTANT: Thus your objective is to maximize:"
1969     f"{C} * (# of agents who choose according to your objective) "
1970     f"- sum of ({d}^i * {k} * abs(message_precision - baseline_precision))\n"
1971     f"where {d} is your discount factor for future cost. "
1972 )
1973 user = (
1974     f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"
1975     f"DECISION HISTORY:\n{history_txt}\n"
1976     f"TASK:"
1977     f"Based on the history and your objective, "
1978     f"choose your desired precision.\n"
1979     f'JSON response: {"message_precision": <number>}\n'
1980 )
1981
1982 Oracle LLM Prompt:
1983
1984 sys = (
1985     f"You will be given "
1986     f"1. an agent profile, "
1987     f"2. information about a car, "
1988     f"3. the true quality of the car, "
1989     f"4. a baseline message and the 'precision' of that message, and "
1990     f"5. a desired precision for the message you output. "
1991     f"(Message precision is the probability (0.5 to 1.0) "
1992     f"that the message leaves the buyer with the correct impression. "
1993     f"For example, if the car is likely GOOD: "
1994     f"a high precision message will likely suggest BUY "
1995     f"and a low precision message will likely suggest NOT BUY. "
1996     f"IMPORTANT: Output a message with the requested precision. "
1997     f"IMPORTANT: The message should be tailored to the profile. "
1998 )
1999 user = (
2000     f"AGENT PROFILE:{profile}\n"
2001     f"TRUE CAR QUALITY: {omega}\n\n"
2002     f"CAR FACTS: {car_facts}\n"
2003     f"BASELINE MESSAGE:{baseline_message}\n"
2004     f"BASELINE MESSAGE precision: {p}\n"
2005     f"DESIRED MESSAGE precision: {q}\n"
2006     f"IMPORTANT: Output a message with the requested precision. "
2007     f"TASK: Write a message (MAX 100 words) with the desired precision."
2008     f'JSON response: {"message": "<message>"}\n'
2009 )

```

E.3 VALIDATION OF LLM BELIEFS AND ORACLE

We now report two further experiments that validate, respectively, the self-reported beliefs of the LLM agents (used in figure 1b) and the precision control of the LLM oracle. Throughout, we treat the LLMs as rational and utility-maximizing but not necessarily Bayesian agents. The experiments are designed to establish two points that the main results rely upon: (1) that an agent’s self-reported posterior genuinely corresponds to the belief governing its choices in state-contingent decision problems—so that the belief measurements of Section 6.1 are behaviorally meaningful—and (2) that the messages produced by the oracle do attain the requested precision.

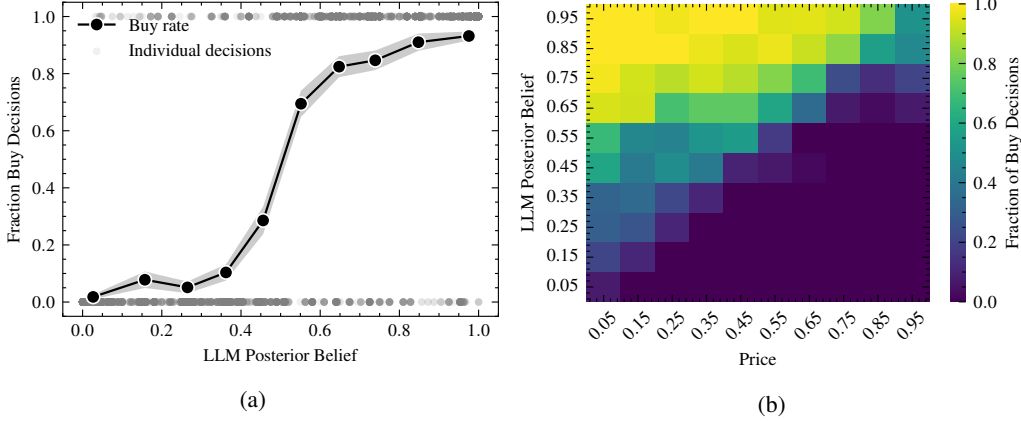


Figure 4: The rate at which the LLM chose a ‘buy’ decision, as a function of (a) its self-reported posterior belief and (b) both its self-reported posterior and the price of buying. In (a) the buy rate switches sharply at the posterior 0.5; in (b) it separates cleanly along the line where price equals belief—both signatures of a utility-maximizing decision maker whose operative belief is the reported posterior.

First, we provide the LLM with a prior and a new signal, obtain its self-reported posterior belief, and then ask it to make a binary “buy/not-buy” decision yielding unit reward if and only if the true state is good. A rational agent should choose “buy” exactly when its subjective belief exceeds 0.5. As figure 4a shows, the LLM’s decisions align sharply with this threshold: it overwhelmingly chooses “buy” when its reported posterior exceeds 0.5 and “not buy” otherwise. The self-reported posterior therefore predicts the subsequent action, which is the property the belief-based analysis requires. Next, we elicit the model’s willingness to pay for a payoff contingent on the true state. For a range of prices c , we ask whether the agent would pay c in exchange for receiving unit utility if the state is good. A rational agent with belief p should accept exactly when $c < p$. Figure 4b plots the fraction of “buy” decisions across posterior–price pairs and reveals a clean separation: acceptance rates are high when $c < p$ and low when $c > p$, exactly as expected of a utility-maximizing decision maker whose operative belief is p . Together, the two experiments establish that the self-reported posterior closely tracks the belief the model actually acts upon.

We validate the oracle’s precision control analogously. After the oracle generates a message of specified precision q , we ask a separate LLM to estimate that message’s precision, yielding $\hat{q}$. We repeat this for $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ across 100 programmatically generated agent profiles with 10 trials per combination, for a total of 6,000 trials. The mean absolute error $|q - \hat{q}|$ was just 0.025, confirming that the realized precision of the oracle’s messages closely follows the requested precision and validating the oracle as a faithful implementation of the signal channel of Section 3.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

We now unpack belief trajectories from a representative instance of the LLM experiment to trace how the non-Bayesian updating of Section 6.1 propagates to the population level. We fix the policy to the analytically optimal one and apply it to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly toward 0, whereas the LLM belief (solid black line) is markedly more “stubborn.” This macro-level trajectory follows directly from the micro-level biases: the public belief resists entering a cascade (NB3) because individual agents overreact to the occasional positive signal that runs counter to the group’s increasingly negative belief (NB2). The trajectory-level evidence thus corroborates, at the level of the full sequence, the single-step belief deviations documented in figure 1b.

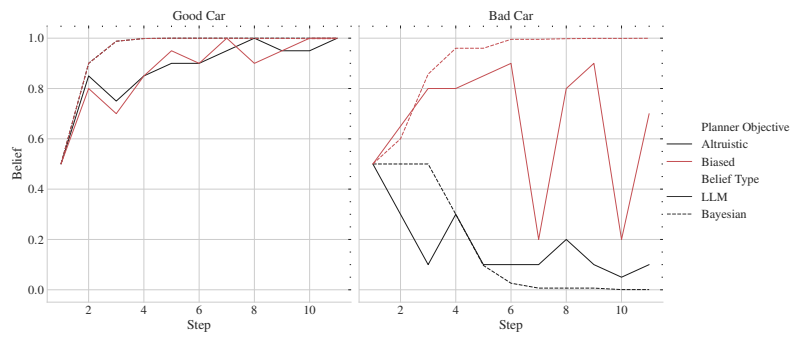


Figure 5: Belief-updating trajectories for the true Bayesian belief and the LLM agent’s reported belief under the analytically optimal policy, across planner objectives and true states of the world. The persistent gap between the Bayesian and LLM trajectories, most pronounced when the car is bad, is the population-level imprint of the single-step non-Bayesian updating of figure 1b.